

10

MEASUREMENT AND SPACE

SURFACE AREA AND VOLUME

The Sydney Opera House was designed by Danish architect Jørn Utzon, with construction taking place between 1959 to 1973 at a cost of $102 million. Its shell design represents a ship under full sail, with each concrete shell being segments of a sphere of radius 75.2 metres. The shells are covered by 1 056 006 white and cream 120 mm square tiles. The Sydney Opera House stands on 1.3 hectares of land, is 183 metres long and 120 metres wide at its widest point.



Chapter outline

	Working mathematically				
10.01 Metric units	U	F		R	C
10.02 Precision and absolute error	U	F		R	C
10.03 Percentage error	U	F		R	C
10.04 Perimeters and areas of composite shapes	U	F	PS	R	
10.05 Areas of quadrilaterals	U	F	PS	R	
10.06 Circumferences and areas of circular shapes	U	F	PS	R	
10.07 Surface area of a prism	U	F	PS	R	
10.08 Surface area of a cylinder	U	F	PS	R	
10.09 Volumes of prisms and cylinders	U	F	PS	R	C

Wordbank

absolute error The greatest possible error of a measurement, equal to half of the precision (smallest unit) on the measuring instrument's scale

capacity The amount of fluid (liquid or gas) in a container

cross-section A 'slice' of a solid, taken across the solid rather than along it

mega- (abbreviated M) A metric prefix meaning one million

micro- (abbreviated μ) A metric prefix meaning one-millionth

precision The smallest unit on the scale of a measuring instrument

prism A solid shape with identical cross-sections and ends with straight sides

sector A region of a circle cut off by 2 radii, shaped like a slice of pizza

surface area The total area of all the faces of a solid shape

Quiz
Wordbank 10

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (8):

10.03 Percentage error

10.06 Circumference of a circle • Area of a circle • Perimeter and area of a sector

10.07 Surface area of a prism • Surface area of prisms

10.08 Surface area of a cylinder

10.09 Volumes of prisms and cylinders

Twig videos (3):

10.02 Decimal places: Photofinish

10.06 Beating the U-boats

10.07 Cubist art

Quizzes (6):

- Wordbank 10
- SkillCheck 10
- Mental skills 10A
- Mental skills 10B
- Language of maths 10
- Test yourself 10

Skillsheets (3):

10.04 What is area?

10.07 Solid shapes

10.09 What is volume?

Worksheets (13):

10.02 Accuracy in measurement 2

10.05 Area ID

10.06 A page of circular shapes • Applications of area • Back-to-front problems • Area and perimeter investigations • Nets of solids

10.07 Surface area 1 • Nets of solids

10.08 Surface area 2 • A page of prisms and cylinders • Applications of area 3

10.09 A page of prisms and cylinders • Volume and capacity

Mind map Surface area and volume

Puzzles (4):

10.06 Carpet talk

10.08 Car song

10.09 Formula matching game

Language of maths Surface area and volume crossword

Technology (1):

10.07, 10.08 Surface area calculator

Spreadsheets (1):

10.07, 10.08 Surface area calculator

Presentation (1):

10.07 Surface area

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- learn the metric prefixes for very small and very large units of measurement
- calculate the absolute error and percentage error of a measurement
- calculate the perimeters and areas of quadrilaterals, circles, sectors and composite shapes
- calculate the surface areas of rectangular and triangular prisms
- calculate the surface areas of prisms and cylinders
- calculate the volumes and capacities of prisms and cylinders

SkillCheck ANSWERS ON P. 587

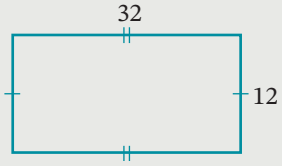
Quiz
SkillCheck 10

10

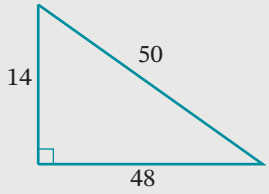
1 The measurements shown on the shapes below are in centimetres. For each shape, find (correct to one decimal place if necessary):

i its perimeter **ii** its area

a



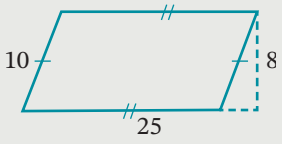
b



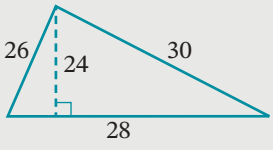
c



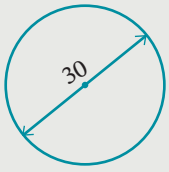
d



e



f



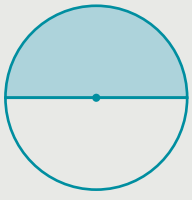
2 Convert:

a 34 m to mm **b** 925 cm to m **c** 1500 m to km

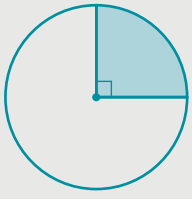
d 3750 mL to L **e** 38 cm to mm **f** 7.5 L to mL

3 What fraction of each circle is shaded?

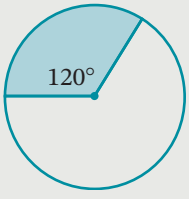
a



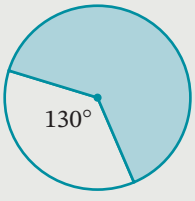
b



c

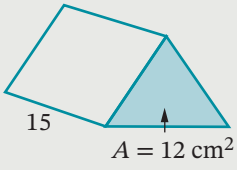


d

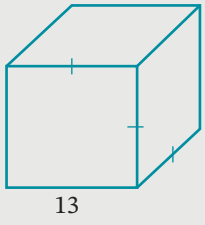


4 Calculate the volume of each prism (all units are in cm).

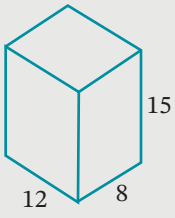
a



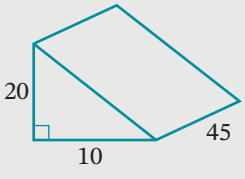
b



c



d



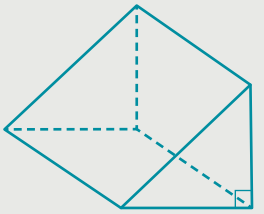
5 For each prism:

i count the number of faces

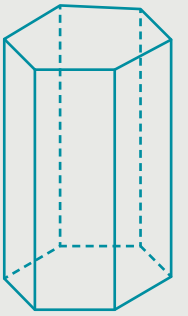
ii name the shapes of the different faces

iii name the prism.

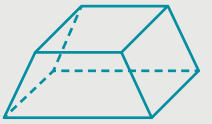
a



b

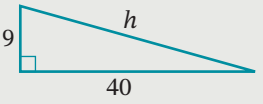


c

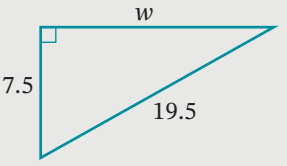


6 Find the value of each variable.

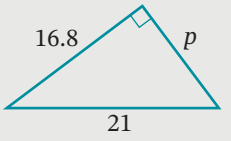
a



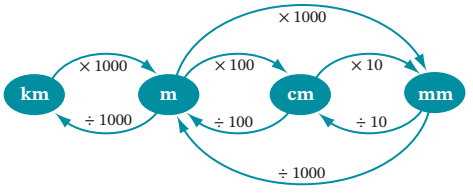
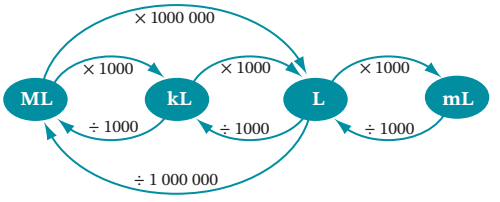
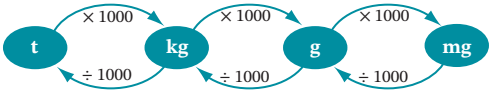
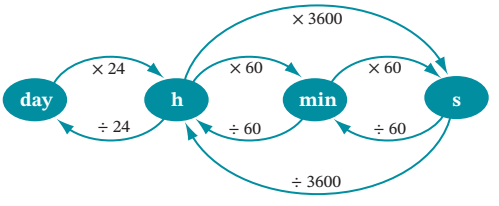
b



c



10.01 Metric units

Length	Capacity
1 cm = 10 mm 1 m = 100 cm = 1000 mm 1 km = 1000 m	1 L = 1000 mL 1 kL = 1000 L 1 ML = 1000 kL = 1000 000 L
	
Mass	**Time**
1 g = 1000 mg 1 kg = 1000 g 1 t = 1000 kg	1 min = 60 s 1 h = 60 min = 3600 s 1 day = 24 h
	

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Other than for time, units of the metric system have prefixes based on powers of 10.

- **kilo-** means 1000 (thousand) or 10^3 — 1 kilogram = 1000 grams
- **mega-** means 1 000 000 (million) or 10^6 — 1 megalitre = 1 000 000 litres
- **centi-** means $\frac{1}{100}$ (hundredth) or 10^{-2} — 1 centimetre = $\frac{1}{100}$ metre
- **milli-** means $\frac{1}{1000}$ (thousandth) or 10^{-3} — 1 millilitre = $\frac{1}{1000}$ litre

Example 1

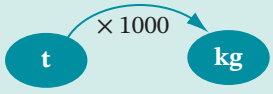
Convert:

a 5.8 t to kg **b** 16 km to cm **c** 23 700 mL to L **d** 1500 min to h

Solution

a $5.8 \text{ t} = 5.8 \times 1000 \text{ kg}$
$= 5800 \text{ kg}$

To convert from large to small units, multiply (×)

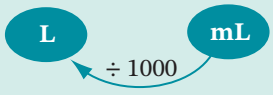


b $16 \text{ km} = 16 \times 1000 \times 100 \text{ cm}$
$= 1\,600\,000 \text{ cm}$

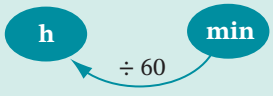


c $23\,700 \text{ mL} = 23\,700 \div 1000 \text{ L}$
$= 23.7 \text{ L}$

To convert from small to large units, divide (÷)



d $1500 \text{ min} = 1500 \div 60 \text{ h}$
$= 25 \text{ h}$



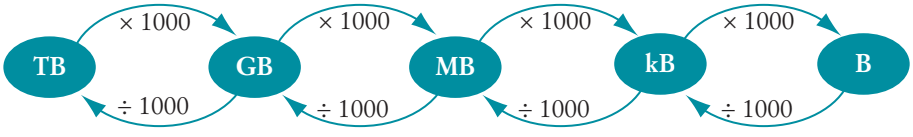
Metric prefixes

This table shows a more detailed list of metric prefixes.

Prefix (abbreviation)	Meaning	Example
pico (p)	10^{-12}	
nano (n)	$10^{-9} = \frac{1}{1000000000}$	nanosecond (ns) = billionth of a second
micro (μ)	$10^{-6} = \frac{1}{1000000}$	microgram (μg) = millionth of a gram
milli (m)	$10^{-3} = \frac{1}{1000}$	millibar (mbar) = thousandth of a bar (air pressure)
centi (c)	$10^{-2} = \frac{1}{100}$	centimetre (cm) = hundredth of a metre
deci (d)	$10^{-1} = \frac{1}{10}$	decibel (dB) = tenth of a bel (sound level)
deca (da)	$10^{1} = 10$	decametre (dam) = ten metres
hecto (h)	$10^{2} = 100$	hectopascal (hPa) = hundred pascals (air pressure)
kilo (k)	$10^{3} = 1000$	kilojoule (kJ) = thousand joules (energy)
mega (M)	$10^{6} = 1\,000\,000$	megahertz (MHz) = million hertz (frequency)
giga (G)	10^{9}	gigalitre (GL) = billion litres (water)
tera (T)	10^{12}	terawatt (TW) = trillion watts (power)

Metric prefixes are used in units of digital (computer) memory. A **byte**, abbreviated B, is the amount of memory used to store one character, such as @, g, #, 7 or ?.

Unit	Relationship	Size
kilobyte (kB)	1 kB = 1000 B	About half a page of text.
megabyte (MB)	1 MB = 1000 kB	A million bytes, about the size of a novel.
gigabyte (GB)	1 GB = 1000 MB	A billion bytes, about 500 photos or 7.5 hours of video.
terabyte (TB)	1 TB = 1000 GB	A trillion bytes, about the size of all of the books in a large library.



Some examples of digital memory sizes:

E-mail (without attachment): 75 kB

Photo: 2 MB

Music file: 4 MB

CD: 750 MB

Streaming a video: 790 MB per hour

DVD: 4.7 GB

Blu-ray disc: 25 GB

Online game: 40 to 300 MB per hour

Example 2

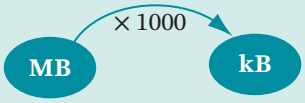
Convert:

a 3.2 MB to kB **b** 147 000 MB to GB **c** 2 TB to GB

Solution

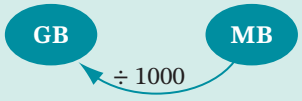
a $3.2 \text{ MB} = 3.2 \times 1000 \text{ kB}$
$= 3200 \text{ kB}$

To convert from large to small units, multiply (×)

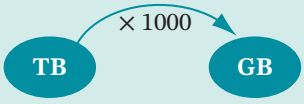


b $147\,000 \text{ MB} = 147\,000 \div 1000 \text{ GB}$
$= 147 \text{ GB}$

To convert from small to large units, divide (÷)



c $2.6 \text{ TB} = 2.6 \times 1000 \text{ GB}$
$= 2600 \text{ GB}$



Very small and very large units

Some very small and very large units of length, distance and time also use metric prefixes.

Length and distance

- One **nanometre** (nm) $= \frac{1}{1\,000\,000\,000}$ m: your fingernails grown 1 nm every second, the diameter of a helium atom is 0.1 nm
- One **micrometre** (μm) $= \frac{1}{1\,000\,000}$ m: used to measure diameters of bacteria
- One **astronomical unit** (AU) = 149 597 871 km: the average distance between the Earth and the Sun, used to measure distances between planets in our solar system
- One **light year** (ly) $= 9.46 \times 10^{12}$ km: the distance light travels in a year, for measuring distances between stars
- One **parsec** (pc) $= 3.0857 \times 10^{13}$ km or 3.26 ly: for measuring distances between stars, the nearest star *Proxima Centauri* is 1.30 parsecs away

Time

- One **nanosecond** (ns) $= \frac{1}{1\,000\,000\,000}$ s: the time it takes electricity to travel along a 30 cm wire
- One **microsecond** (μs) $= \frac{1}{1\,000\,000}$ s
- One **millisecond** (ms) $= \frac{1}{1\,000}$ s: the duration of the flash on a camera
- One **decade** = 10 years

- One **century** = 100 years
- One **millennium** = 1000 years
- One **mega-annum** (Ma) = 1 000 000 years: used in geology and paleontology, the *Tyrannosaurus Rex* dinosaur lived 65 Ma ago
- One **giga-annum** (Ga) = 1 000 000 000 years: the Earth formed 4.57 Ga ago

Example 3

Convert:

a 0.73 s to μs

b 25 610 000 km to AU

c 4.57 Ma to years

d 1 920 000 nm to m

Solution

a $0.73 \text{ s} = 0.73 \times 1\,000\,000 \text{ μs}$
$= 730\,000 \text{ μs}$

To convert from large to small units, multiply (×)

b $25\,610\,000 \text{ km} = 25\,610\,000 \div 149\,597\,871 \text{ AU}$
$= 0.17119...$
$\approx 0.17 \text{ AU}$

To convert from small to large units, divide (÷)

c $4.57 \text{ Ma} = 4.57 \times 1\,000\,000 \text{ years}$
$= 4\,570\,000 \text{ years}$

d $1\,920\,000 \text{ nm} = 1\,920\,000 \div 1\,000\,000\,000 \text{ m}$
$= 1.92 \times 10^{-3} \text{ m}$
$= 0.001\,92 \text{ m}$

EXERCISE 10.01 ANSWERS ON P. 587

Metric units U F R C

1 What metric unit would you use for each measurement? R C

a Width of a computer screen

b Mass of an apple

c Distance between Newcastle and Forster

d Capacity of a car's fuel tank

e Amount of water in a swimming pool

f Time taken to fly overseas

EXAMPLE 1

2 Convert:

a 750 g to mg

b 36 kL to L

c 67 kg to g

d 1.5 L to mL

e 6.7 cm to mm

f 3.5 km to m

g 9.4 t to kg

h 3.75 g to mg

i 4.25 m to cm

j 0.2 kg to g

k 26.2 L to mL

l 0.3 km to m

m 3 kL to mL

n 4 days to min

o 15 hr to s

p 2.7 km to cm

q 5 kg to mg

r 0.7 m to mm

s 29.375 t to g

t 18.14 km to mm

u 3.27 g to mg

Foundation Standard Complex

3 How many mL in 18.9 kL? Select the correct answer **A**, **B**, **C** or **D**.

A 1 890 000 **B** 18 900 000 **C** 189 000 **D** 18 900

4 Select the most appropriate answer **A**, **B** or **C** for each measurement.

a height of a door

A 20 cm **B** 200 cm **C** 20 m

b capacity of a soft drink can

A 37.5 mL **B** 375 mL **C** 3.5 L

c time to run 100 metres

A 12.3 s **B** 34.3 s **C** 50.3 s

d height of Sydney Tower

A 30.5 m **B** 3050 m **C** 305 m

e mass of a family car

A 100 kg **B** 1000 kg **C** 10 000 kg

5 Convert:

a	8000 L to kL	**b**	90 mm to cm	**c**	180 s to min
d	240 min to h	**e**	375 mL to L	**f**	14 700 kg to t
g	8600 mg to g	**h**	750 cm to m	**i**	223 mm to cm
j	800 g to kg	**k**	125 L to kL	**l**	3.5 kg to t
m	17 000 000 mg to kg	**n**	7200 min to days	**o**	45 000 000 g to t
p	144 000 s to h	**q**	86 400 s to days	**r**	2500 cm to km
s	75 000 mL to kL	**t**	34 700 mm to m	**u**	628 L to ML

6 How many km in 408 m? Select **A**, **B**, **C** or **D**.

A 4.08 **B** 0.408 **C** 0.0408 **D** 0.004 08

7 The capacity of Sydney Harbour is approximately a quarter of Warragamba Dam's capacity. If the capacity of Warragamba dam is 2013 GL, what is the capacity of Sydney Harbour in litres?

8 Convert: R C

a	4.8 MB to kB	**b**	2.1 TB to GB	**c**	8.5 GB to MB
d	10.8 MB to B	**e**	2910 B to kB	**f**	740 MB to GB
g	1050 kB to MB	**h**	5900 GB to TB	**i**	0.94 kB to MB
j	57 GB to MB	**k**	0.43 GB to kB	**l**	21 TB to GB

EXAMPLE 2

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9 Ritu made a video call lasting $1\frac{1}{4}$ hours. If it used 3MB/min, how much data did the call take?

10 Calculate the number of MP3 files of average size 5 MB that can be stored on a 64 GB MP3 player. R

EXAMPLE 3

11 Convert, correct to 2 significant figures: R C

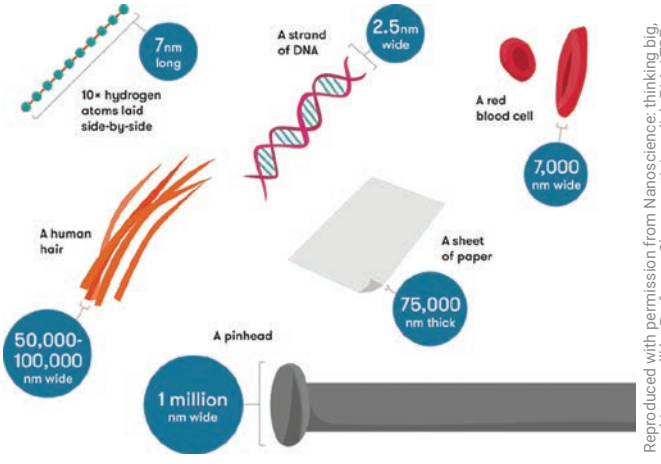
a	3.5 AU to km	**b**	6000 ms to s	**c**	7.5 centuries to years
d	12 000 000 nm to m	**e**	0.64 Gm to m	**f**	2.1 ly to km
g	0.07 s to ms	**h**	0.25 Ma to years	**i**	4 800 000 μs to s
j	5 290 000 km to Mm	**k**	0.2 pc to km	**l**	6 mm to μm
m	0.94 Ma to millennia	**n**	849 years to centuries	**o**	8×10^{14} km to ly
p	0.000 008 cm to nm	**q**	20 ly to pc	**r**	156 millennia to Ma

Did you know?

Nanotechnology

One nanometre is equal to one billionth of a metre. This length is so small that it is used to measure cells and bacteria. Nanotechnology is the study of extremely small matter of size 1 to 100 nanometres, also known as the nanoscale. At this scale, matter can possess unusual physical, chemical and biological properties, different to single atom and molecule properties, as nanostructured matter is stronger or contains different magnetic properties in comparison to other forms or sizes of similar material. Such materials can be more chemically reactive, reflect light better or change colour, as their size or form changes.

To understand the comparative size of a nanometre, some examples include:



Reproduced with permission from Nanoscience: thinking big, working small' by Professor Chennupati Jagadish FAA ATSE and Dr Amanda Barnard, Australian Academy of Science, https://www.science.org.au/curious/nanoscience

Sunscreen, clothing, furniture and computers all incorporate nanotechnology. Research how this is done and find other objects that exist using nanotechnology.

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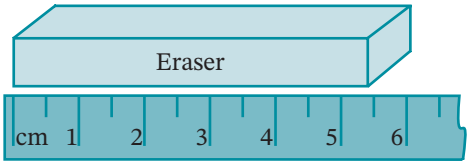
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Precision and absolute error

10.02

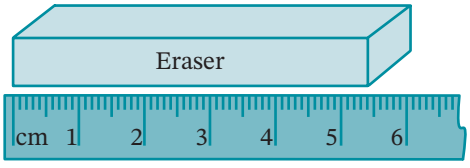
All measurements are only approximations. No measurement is ever **exact**.

For example, we might measure this eraser to be **5.5 cm** long, because the scale on the ruler measures to the nearest 0.5 cm.



If we used a ruler with a more precise scale, such as 0.1 cm markings, then we may find the length to be **5.4 cm**.

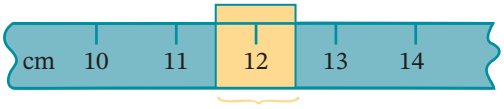
If we used an instrument such as a micrometer that measured to the nearest 0.01 cm, we may find that the length is **5.41 cm**.



Notice that we can always find a more accurate measurement using a more precise measuring instrument. So all measurements are approximations.

A measuring instrument such as a ruler has **limits of accuracy** due to its level of **precision**. 'Accuracy' means how close a measured value is to the true value, while 'precision' means how fine the scale is on the measuring instrument. In measurement, 'error' refers to the difference between a measured value and the true value.

For example, this ruler is marked in centimetres, so any length measured with it can only be given to the nearest centimetre. This ruler has a **precision** of 1 cm.



Any measurement in the shaded region should be recorded as 12 cm. The measured length is 12 cm, but the actual length is 12 ± 0.5 cm, meaning '12 centimetres, give or take 0.5 centimetres' or '11.5 to 12.5 cm'.

The **absolute error** of this ruler is **0.5 cm**, which represents the greatest possible error the actual measurement can differ from the recorded measurement (12 cm), the furthest your measurement could be wrong.

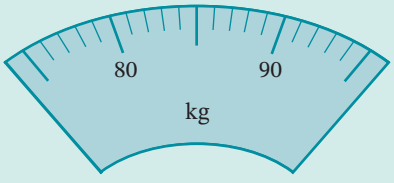
Absolute error of a measurement

The **absolute error** of a measurement is 0.5 (half) of the precision of the measuring instrument's scale.

Example 4

Find the absolute error of each measuring scale.

a



b



Worksheet
Accuracy in measurement 2

Video
Decimal places: Photofinish

Solution

a The precision (size of one unit) of the scale is 1 kg.

The absolute error is $0.5 \times 1 \text{ kg} = 0.5 \text{ kg}$

b The precision of the scale is 5 mL.

The absolute error is $0.5 \times 5 \text{ mL} = 2.5 \text{ mL}$.

EXERCISE 10.02 ANSWERS ON P. 588

Precision and absolute error U F R C

EXAMPLE 4

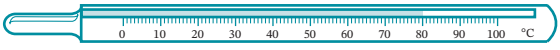
1 For each measuring instrument, state: R C

i its precision **ii** its absolute error.

a



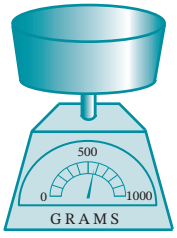
b



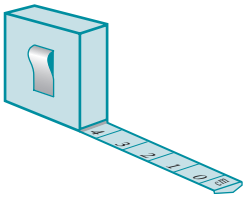
c



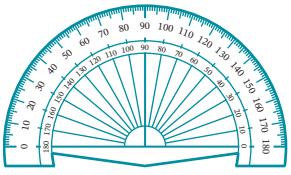
d



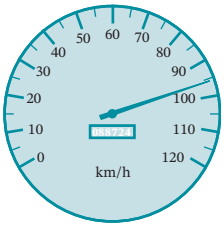
e



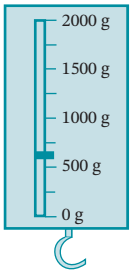
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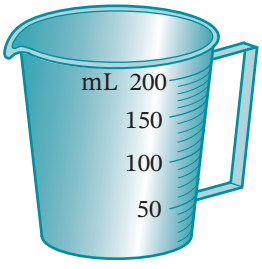
g



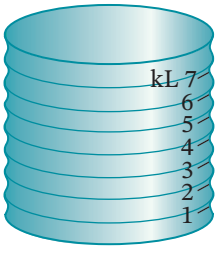
h



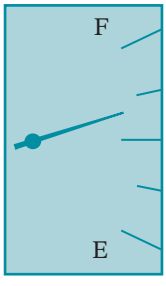
i



j



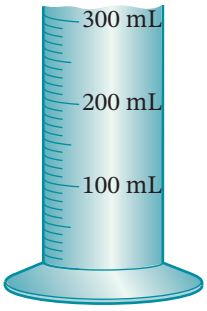
k



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2 What is the absolute error of this measuring cylinder?
Select the correct answer **A**, **B**, **C** or **D**.



A 0.5 mL

B 10 mL

C 20 mL

D 5 mL

3 Measuring with a ruler, Gemma correctly gives a measurement as 26.5 to 27.5 cm. R C

a What is the precision of the ruler?

b What is the absolute error of this measurement?

c Using a ruler marked in millimetres, Emily gives the same measurement as 26.8 cm. What is the absolute error of this measurement?

d What is the range within which this measurement must lie?

Percentage error

10.03

All measurements are approximate. Whether we are using a ruler to measure distance, weighing scales for recipe ingredients or using a thermometer to measure temperature, there is some error in every measurement we make. An absolute error of 1 metre may not be much when measuring the length of a bridge, but it is huge when measuring the length of a bathroom. To understand how big the errors are when something is measured and be able to compare errors in measurement, we can calculate **percentage error**, the absolute error as a percentage of the measurement.

Percentage error of a measurement

$$\text{Percentage error} = \frac{\text{absolute error}}{\text{measurement}} \times 100\%$$

The smaller the percentage error, the more accurate the measurement, the closer it is to the actual value.

Example 5

Sydney Tower is 309 metres tall.

Find the percentage error of this measurement, correct to 2 decimal places.


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Solution

Absolute error $= 0.5 \times 1$ m
$= 0.5$ m

Precision $= 1$ m

Percentage error $= \frac{0.5}{309} \times 100$
$= 0.1618123 \ldots$
$\approx 0.16\ \%$

EXERCISE 10.03 ANSWERS ON P. 588

Percentage error U F R C

1 Copy and complete this table. R C

	Measurement	Precision	Absolute error
a	60 cm		
b	42 seconds		
c	10.5 mL		
d	150 cm (to the nearest cm)		
e	150 cm (to the nearest 10 cm)		

2 Find the absolute error of each measurement.

a 150 mL
b 3640 mm
c 5.2 g
d 200 g (to the nearest g)
e 200 g (to the nearest 10 g)
f 200 g (to the nearest 100 g)

EXAMPLE 5

3 Dinika's car weighs 1200 kg, to the nearest 100 kg. Calculate the absolute error of this measurement and then its percentage error, correct to 2 decimal places.

4 The Sydney Harbour Bridge spans 503 metres.

a Find the absolute error of this measurement.

b Hence calculate the percentage error correct to 3 decimal places.

5 **a** Q1 Tower on the Gold Coast in Queensland is 322.5 metres high. Calculate its percentage error, correct to 3 decimal places.

b Which measurement is more accurate, the length of the Sydney Harbour Bridge from Question 4, or the height of the Q1 Tower from part **a**? R C



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6 The longest non-stop passenger flight was from Perth to London in 2022, a distance of 14 498 km. Calculate the percentage error of this measurement, correct to 5 decimal places.

7 The size of an Olympic swimming pool is 50 metres in length. Given it is measured to the nearest metre, what is the percentage error?

8 **a** A standard cricket pitch, between the stumps, is 20.12 metres long. Determine the percentage error of this measurement, correct to 3 decimal places.

b Which measurement is more accurate, the length of the pool from Question 7, or the length of the cricket pitch from part a? R C

9 Jenny was filling her car with petrol and told Claire that she had put 44 L of petrol in. Claire looked at the fuel pump and replied that 44.0 L of fuel had been added. Explain which measurement is more accurate. R C

10 The side length of a cube was measured. Find the percentage error of each measurement.

a 4 cm **b** 4.0 cm

Perimeters and areas of composite shapes

10.04

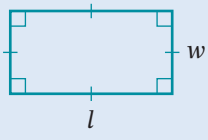
The **perimeter** of a shape is the distance around the shape, the sum of the lengths of the sides of the shape.

The **area** of a shape is the amount of surface covered by the shape, measured in **square units**.

Skillsheet
What is area?

Areas of rectangles and triangles

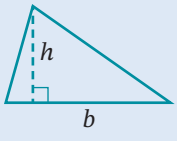
Rectangle



$A = \text{length} \times \text{width}$

$A = lw$

Triangle



$A = \frac{1}{2} \times \text{base} \times \text{perpendicular height}$

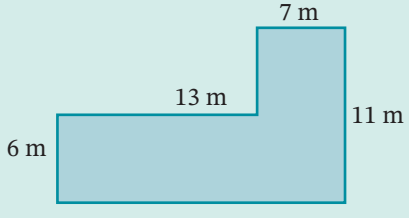
$A = \frac{1}{2}bh$

Example 6

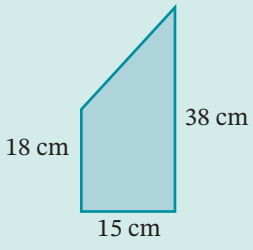
For each **composite shape**, find:

i its perimeter **ii** its area.

a



b



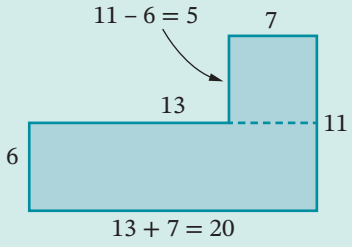
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Solution

a **i** Find any unknown lengths on the shape.

Perimeter = 6 + 13 + 5 + 7 + 11 + 20

= 62 m



ii Area = (7 × 5) + (20 × 6) Sum of areas of 2 rectangles

= 155 m²

b **i** Divide the shape into a rectangle and a triangle, and find the unknown length, h, using Pythagoras' theorem.

$h^2 = 15^2 + 20^2$

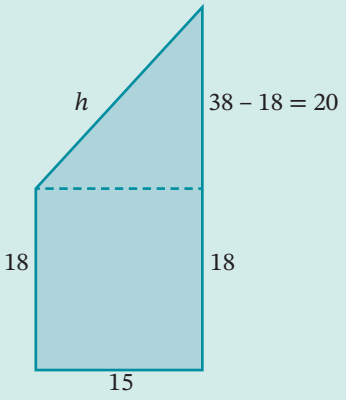
$= 625$

$h = \sqrt{625}$

$= 25$ cm

Perimeter = 18 + 25 + 38 + 15

= 96 cm



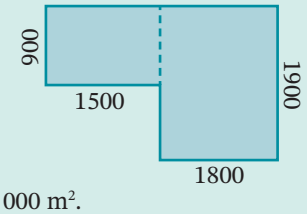
ii Area = $(\frac{1}{2} \times 15 \times 20) + (15 \times 18)$ Area of triangle + Area of rectangle

= 420 cm²

Example 7

This is the plan of a farm up for sale.

All measurements are in metres.



a Find the area of the farm.

b Express the area in square kilometres.

c Find the area of the property in **hectares**, where 1 ha = 10 000 m².

Solution

a Area = 900 × 1500 + 1900 × 1800

= 4 770 000 m²

b Since 1 km = 1000 m, 1 km² = 1000² m² = 1 000 000 m²

∴ Area = 4 770 000 ÷ 1 000 000

= 4.77 km²

c 1 ha = 10 000 m²

∴ Area = 4 770 000 ÷ 10 000

= 477 ha

EXERCISE 10.04 ANSWERS ON P. 588

Perimeters and areas of composite shapes U F PS R

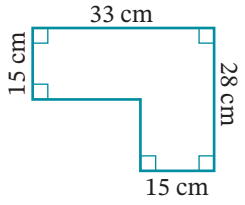
EXAMPLE 6

10.04

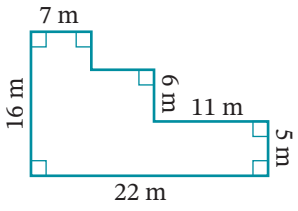
1 For each composite shape, find: PS R

i its perimeter **ii** its area.

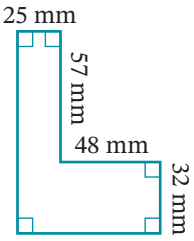
a



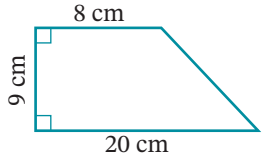
b



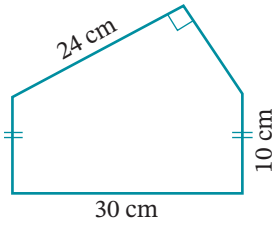
c



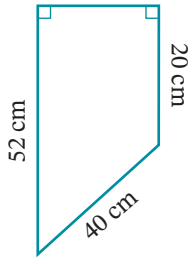
d



e



f

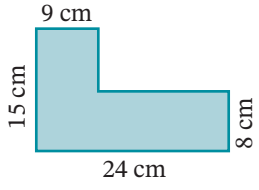


2 The length of a rectangular block of land is 112 m and its perimeter is 334 m.
How wide is the block? Select the correct answer **A**, **B**, **C** or **D**. R

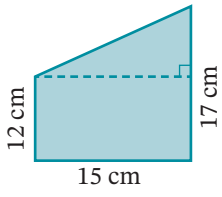
A 222 m **B** 110 m **C** 55 m **D** 224 m

3 Find the area of each composite shape. PS R

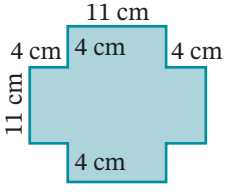
a



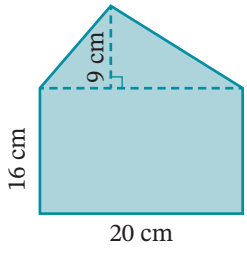
b



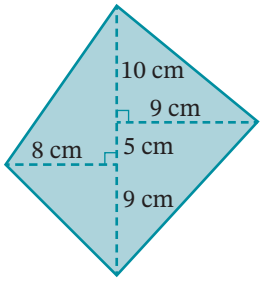
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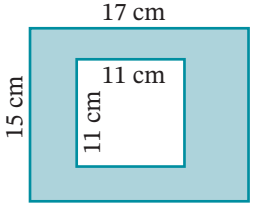
d



e

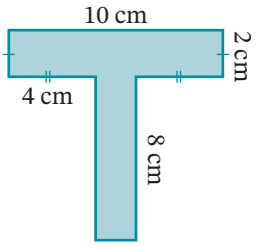


f

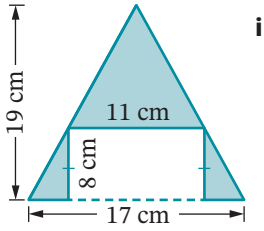


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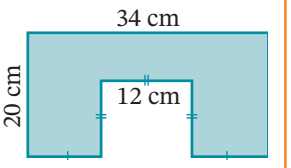
g



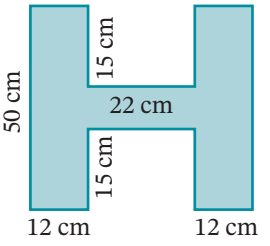
h



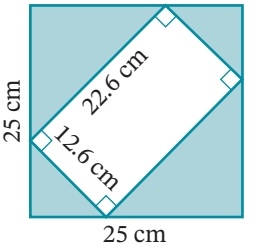
i



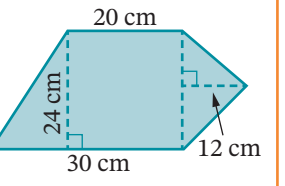
j



k



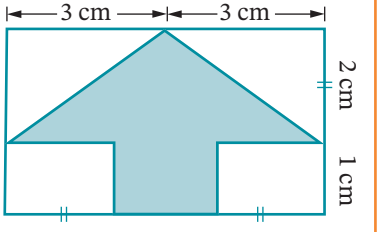
l



4 What is the area of the shaded arrow?
Select **A**, **B**, **C** or **D**.

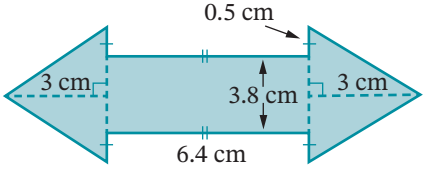
A 14 cm² **B** 8 cm²

C 9 cm² **D** 12 cm²

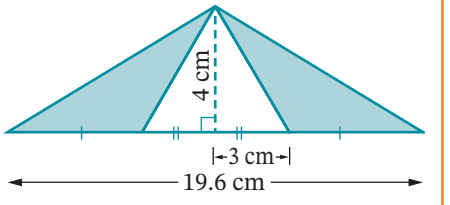


5 Find each shaded area. PS R

a



b

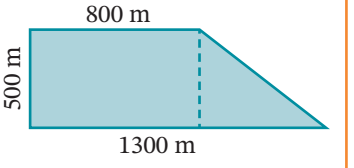


EXAMPLE 7

6 **a** Find the area of the large block of land shown in the diagram (in m²).

b Express the area of the land in:

i km² **ii** hectares



7 The area of a large rectangular property is 8.6 ha. Calculate the width if its length is 400 m. Select **A**, **B**, **C** or **D**.

A 21.5 m **B** 215 m **C** 2150 m **D** 0.0215 km

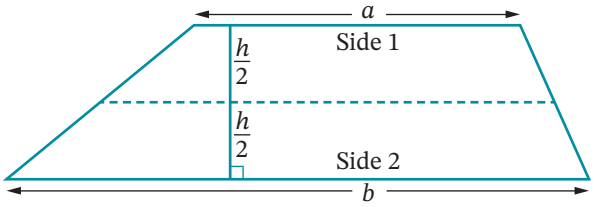
Foundation Standard Complex

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Investigation

Area of a trapezium, kite and rhombus

1 Draw 2 copies of this trapezium onto a sheet of paper. Draw a dotted line, parallel to the sides labelled a and b, halfway down the height of each trapezium and label the sides as shown.



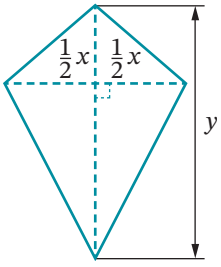
2 Cut out one trapezium and cut along the dotted line to make 2 smaller trapeziums. Join the pieces to make a long parallelogram.

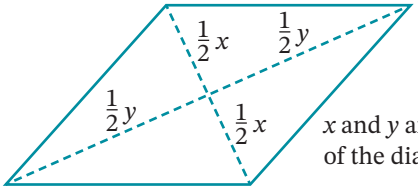
3 By measuring the height and base of the parallelogram to the nearest millimetre, find its area.

4 By comparing the measurements of the parallelogram to the measurements of the original trapezium, suggest a general formula for finding the area of any trapezium. Check your answer with your teacher.

5 Cut out the original trapezium and paste it and the parallelogram into your book.

6 Draw 2 copies of the kite and rhombus on a sheet of paper. Draw diagonals of length x and y and label them as shown. Notice that they cross at right angles. For the kite, one diagonal (x) is bisected by the other (y), while for the rhombus, *both* diagonals bisect each other.





x and y are the lengths of the diagonals

7 Cut out one of your kites and rhombuses and cut along their diagonals to make 4 triangles each. Join each set of triangles to make a rectangle.

8 By measuring the sides of each rectangle to the nearest millimetre, find the area of each one.

9 By comparing the measurements of the kite to the measurements of the rectangle formed from it, suggest a general formula for finding the area of any kite. Check your answer with your teacher.

10 Is a rhombus a special type of kite?

11 Suggest a general formula for finding the area of any rhombus. Check your answer with your teacher.

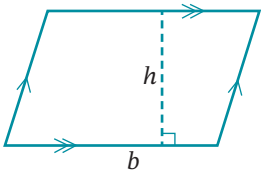
12 Cut out the original kite and rhombus and paste all shapes into your book.

10.05 Areas of quadrilaterals

Worksheet
Area ID

Areas of special quadrilaterals

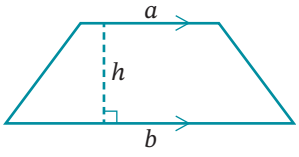
Parallelogram



Area = base × perpendicular height

$A = bh$

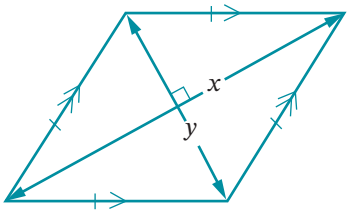
Trapezium



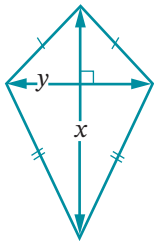
Area $= \frac{1}{2} \times$ (sum of parallel sides) × perpendicular height

$A = \frac{1}{2}(a + b)h$

Rhombus



Kite



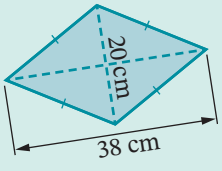
Area $= \frac{1}{2} \times$ (product of diagonals)

$A = \frac{1}{2}xy$

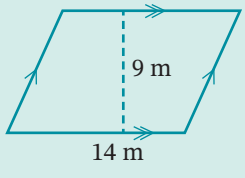
Example 8

Find the area of each quadrilateral.

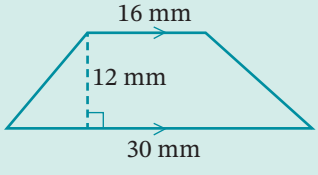
a



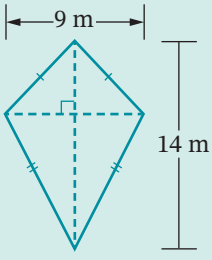
b



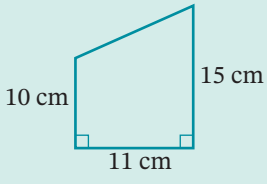
c



d



e



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Solution

a Area of rhombus $= \frac{1}{2} \times 38 \times 20$ $\qquad A = \frac{1}{2}xy$

$= 380 \text{ cm}^2$

b Area of parallelogram $= 14 \times 9$ $\qquad A = bh$

$= 126 \text{ m}^2$

c Area of trapezium $= \frac{1}{2} \times (16 + 30) \times 12$ $\qquad A = \frac{1}{2}(a + b)h$

$= 276 \text{ mm}^2$

d Area of kite $= \frac{1}{2} \times 9 \times 14$ $\qquad A = \frac{1}{2}xy$

$= 63 \text{ m}^2$

e Area of trapezium $= \frac{1}{2} \times (10 + 15) \times 11$ $\qquad A = \frac{1}{2}(a + b)h$

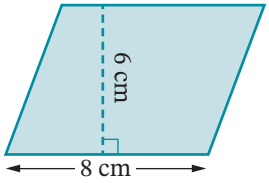
$= 137.5 \text{ cm}^2$

EXERCISE 10.05 ANSWERS ON P. 588

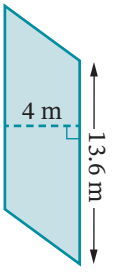
Areas of quadrilaterals U F PS R

1 Find the shaded area of each shape.

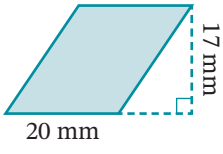
a



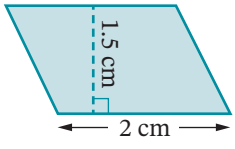
b



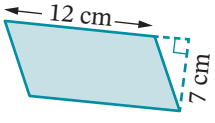
c



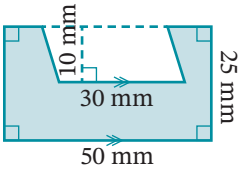
d



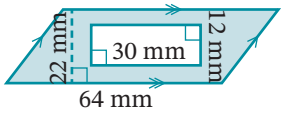
e



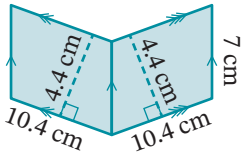
f



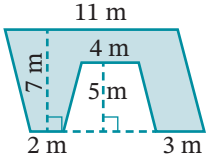
g



h

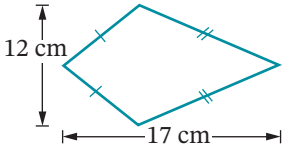


i

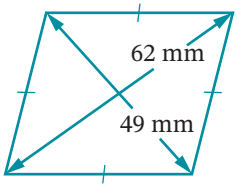


2 Find the area of each shape.

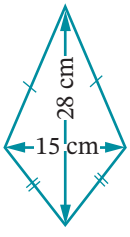
a



b

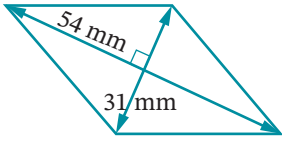


c

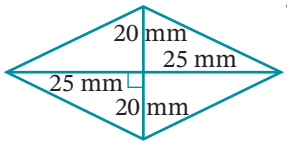


Foundation | Standard | Complex

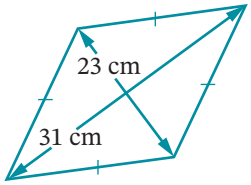
d



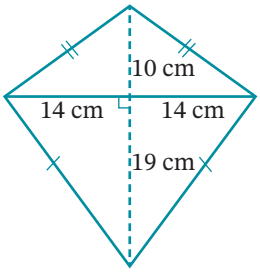
e



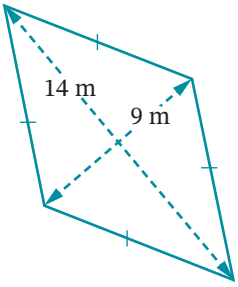
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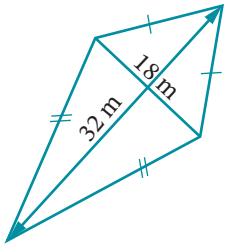
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h

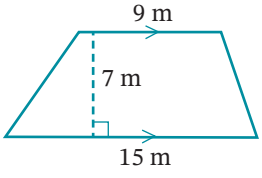


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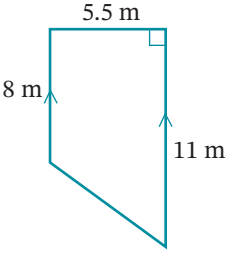


3 Find the area of each shape.

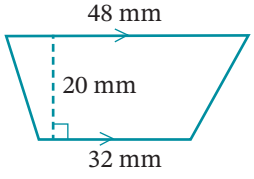
a



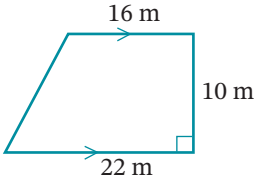
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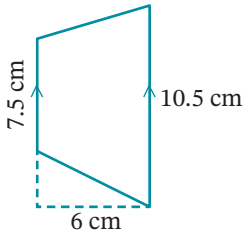
c



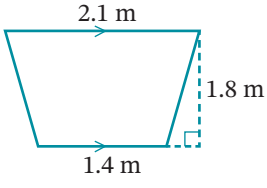
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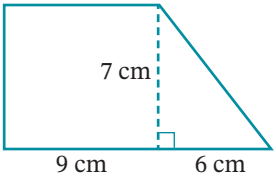
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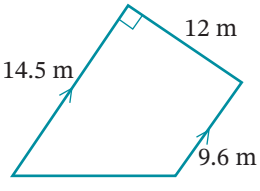
f



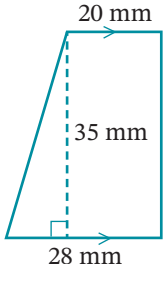
g



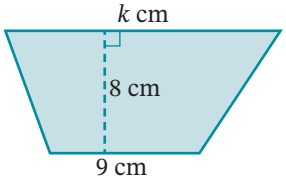
h



i



4 The area of the trapezium is 96 cm^2. What is the length of the missing side? Select the correct answer **A**, **B**, **C** or **D**.



A 10 cm **B** 15 cm

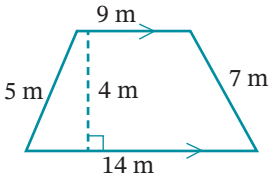
C 17 cm **D** 12 cm

Foundation Standard Complex

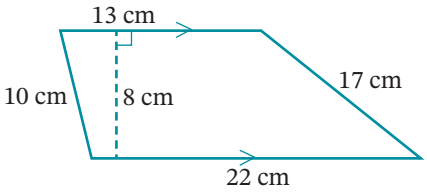
9780170479370

5 Find the area of each shape.

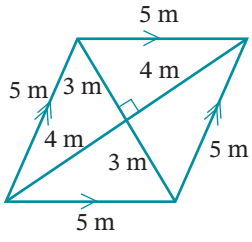
a



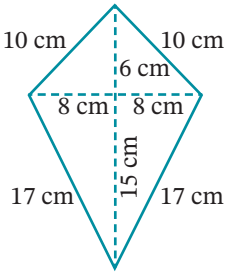
b



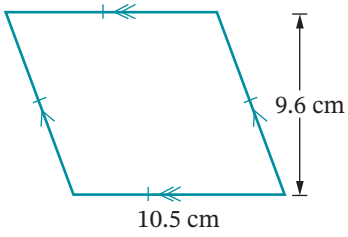
c



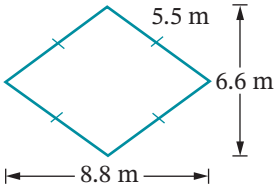
d



e

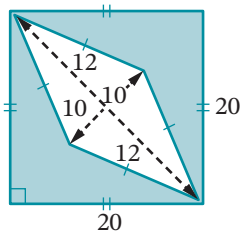


f

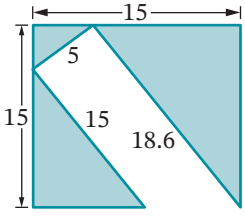


6 Calculate each shaded area (all measurements are in metres). P R

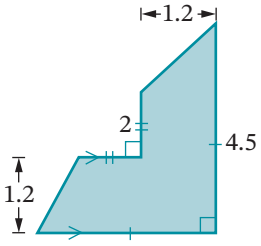
a



b



c



Mental skills 10A: Maths without calculators ANSWERS ON P. 588

Finding 10%, 20% and 5%

To find 10% or $\frac{1}{10}$ of a number, simply divide the number by 10 by moving the decimal point one place to the left.

1 Study each example.

a $10\% \times 150 = 15\overset{\curvearrowleft}{0}. = 15$

b $10\% \times \$1256.80 = \$125\overset{\curvearrowleft}{6}.8 = \$125.68$

c $10\% \times 4917 = 491\overset{\curvearrowleft}{7}. = 491.7$

d $10\% \times \$48.55 = \$4\overset{\curvearrowleft}{8}.55 = \$4.885$

Quiz
Mental skills 10A

Foundation Standard Complex

2 Now find 10% of each amount.

a	190	**b**	$75	**c**	875	**d**	$202
e	$37.60	**f**	400	**g**	$9.25	**h**	896
i	$2700	**j**	$3.80	**k**	$1527.60	**l**	$72.50
m	3154	**n**	$10.70	**o**	426	**p**	$24 317.60

20% is 10% doubled, so to find 20% of a number, first find 10% then double it.

3 Study each example.

a
$20\% \times 700$
$10\% \times 700 = 70$
$\therefore 20\% \times 700 = 70 \times 2$
$= 140$

b
$20\% \times \$876$
$10\% \times \$876 = \$87.60$
$\therefore 20\% \times \$876 = \$87.60 \times 2$
$= \$175.20$

c
$20\% \times 325$
$10\% \times 325 = 32.5$
$\therefore 20\% \times 325 = 32.5 \times 2$
$= 65$

d
$20\% \times \$38.50$
$10\% \times \$38.50 = \$3.85$
$\therefore 20\% \times \$38.50 = \$3.85 \times 2$
$= \$7.70$

4 Now find 20% of each amount.

a	50	**b**	620	**c**	$2450	**d**	$8.60
e	72	**f**	$12 700	**g**	390	**h**	$5.80
i	$45	**j**	$84	**k**	$4600	**l**	320

5% is half of 10%, so to find 5% of a number, first find 10% then divide it by 2.

5 Study each example.

a
$5\% \times 180$
$10\% \times 180 = 18$
$\therefore 5\% \times 180 = 18 \div 2$
$= 9$

b
$5\% \times \$76$
$10\% \times \$76 = \$7.60$
$\therefore 5\% \times \$76 = \$7.60 \div 2$
$= \$3.80$

c
$5\% \times 120$
$10\% \times 120 = 12$
$\therefore 5\% \times 120 = 12 \div 2$
$= 6$

d
$5\% \times \$142.20$
$10\% \times \$142.20 = \$14.22$
$\therefore 5\% \times \$142.20 = \$14.22 \div 2$
$= \$7.11$

6 Now find 5% of each amount.

a	2000	**b**	$12	**c**	50	**d**	$27
e	$36.80	**f**	$84	**g**	800	**h**	130
i	$9.60	**j**	$138	**k**	$72	**l**	840

Circumferences and areas of circular shapes

10.06

The **circumference** of a circle is found by multiplying its **diameter** by a special number called **pi** (pronounced 'pie'), represented by the Greek letter π.

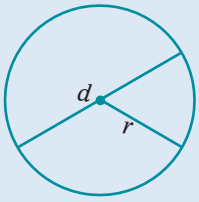
The **area** of a circle is found by multiplying π by the **radius squared.**

Circumference of a circle

The **circumference** (**perimeter**) **of a circle** is:

$C = \pi \times \text{diameter}$ or $C = 2 \times \pi \times \text{radius}$

$C = \pi d$ or $C = 2\pi r$



Area of a circle

The **area of a circle** is:

$$A = \pi \times (\text{radius})^2$$

$$A = \pi r^2$$

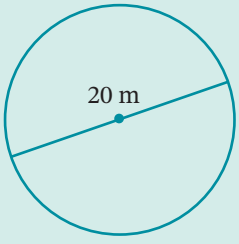
$\pi = 3.141\ 592\ 653\ 897\ 93...$ and can be found by pressing the π key on your calculator. It has no exact decimal or fraction value, so like a surd such as $\sqrt{2}$, it is an **irrational number**.

Example 9

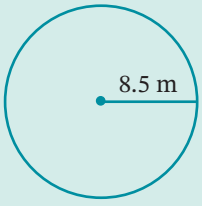
For each circle, calculate correct to 2 decimal places:

i its circumference **ii** its area.

a



b



Solution

a i $C = \pi \times 20$ $C = \pi d$

$= 62.8318...$

≈ 62.83 m

ii $A = \pi \times 10^2$ $A = \pi r^2$

$= 314.1592...$

≈ 314.16 m^2

b i $C = 2 \times \pi \times 8.5$ $C = 2\pi r$

$= 53.4070...$

≈ 53.41 m

ii $A = \pi \times 8.5^2$ $A = \pi r^2$

$= 226.9800...$

≈ 226.98 m^2

Worksheets
A page of circular shapes
Applications of area
Back-to-front problems
Area and perimeter investigations

Puzzle
Carpet talk

Videos
Circumference of a circle
Area of a circle
Beating the U-boats

Video
Perimeter and area of a sector

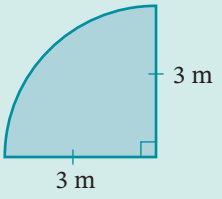
Example 10

For each sector, calculate correct to one decimal place:

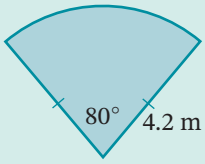
i its perimeter **ii** its area.

A **sector** is a fraction of a circle 'cut' along 2 radii, like a pizza slice.

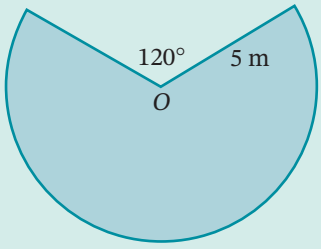
a



b



c



Solution

a i This sector is a **quadrant**, a quarter of a circle. $\frac{1}{4} \times$ circumference + radius + radius

$$\text{Perimeter} = \frac{1}{4} \times 2 \times \pi \times 3 + 3 + 3$$
$$= 10.71238...$$
$$= 10.7 \text{ m}$$

ii $\text{Area} = \frac{1}{4} \times \pi \times 3^2$ $\frac{1}{4} \times$ area of circle

$$= 7.06858...$$
$$= 7.1 \text{ m}^2$$

b i $\text{Perimeter} = \frac{80}{360} \times 2 \times \pi \times 4.2 + 4.2 + 4.2$ $\frac{80}{360} \times$ circumference + radius + radius

$$= 14.26430...$$
$$= 14.3 \text{ m}$$

There are 360° in a circle, but a sector is a fraction of a circle.

ii $\text{Area} = \frac{80}{360} \times \pi \times 4.2^2$ $\frac{80}{360} \times$ area of circle

$$= 12.31504...$$
$$= 12.3 \text{ m}^2$$

c i $\text{Sector angle} = 360° - 120°$

$$= 240°$$

$$\text{Perimeter} = \frac{240}{360} \times 2 \times \pi \times 5 + 5 + 5$$
$$= 30.94395...$$
$$= 30.9 \text{ m}$$

ii $\text{Area of sector} = \frac{240}{360} \times \pi \times 5^2$

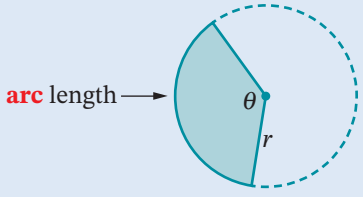
$$= 52.35987...$$
$$= 52.4 \text{ m}^2$$

Arc length, perimeter and area of a sector

For a sector with angle θ:

- Arc length $= \frac{\theta}{360} \times 2\pi r$
- Perimeter of the sector $= \frac{\theta}{360} \times 2\pi r + r + r$

  (arc length + radius + radius)
- Area of the sector $= \frac{\theta}{360} \times \pi r^2$

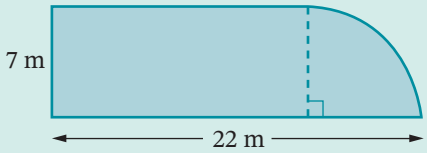


9780170479370

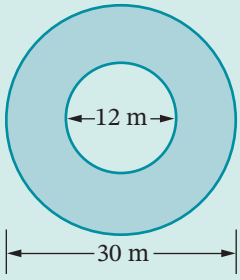
Example 11

Find, correct to one decimal place, the area of each shape.

a



b



Solution

a The shape is made up of a rectangle and a quadrant.

Radius of quadrant $= 7$ m

Length of rectangle $= 22 - 7$

$= 15$ m

Area of shape = area of rectangle + quadrant

$= 15 \times 7 + \frac{1}{4} \times \pi \times 7^2$

$= 143.4845...$

≈ 143.5 m^2

b This ring shape is called an **annulus**, it is the area enclosed by 2 circles with the same centre.

Radius of large circle $= \frac{1}{2} \times 30$ m

$= 15$ m

Radius of small circle $= \frac{1}{2} \times 12$ m

$= 6$ m

Shaded area = large circle – small circle

$= \pi \times 15^2 - \pi \times 6^2$

$= 593.7610...$

≈ 593.8 m^2

EXERCISE 10.06 ANSWERS ON P. 588

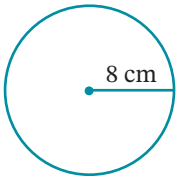
Circumferences and areas of circular shapes U F PS R

EXAMPLE 9

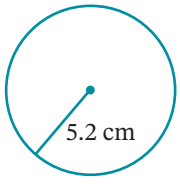
1 For each circle, calculate correct to 2 decimal places:

i its circumference **ii** its area.

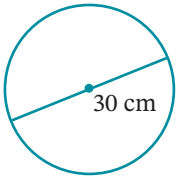
a



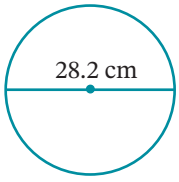
b



c



d



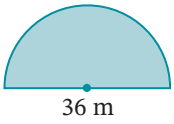
Foundation Standard Complex

EXAMPLE 10

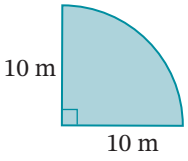
2 For each sector, calculate correct to one decimal place:

i its perimeter **ii** its area.

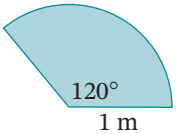
a



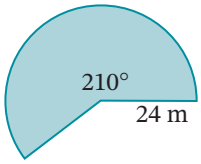
b



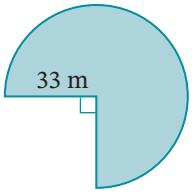
c



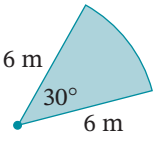
d



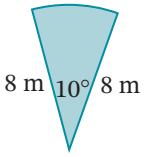
e



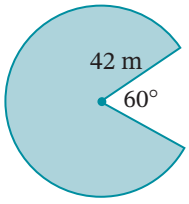
f



g



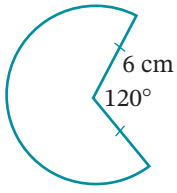
h



3 Find the perimeter of this sector. Select the correct answer **A**, **B**, **C** or **D**.

A 12.6 cm **B** 18.3 cm

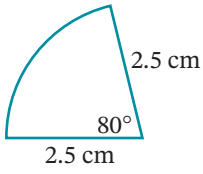
C 24.6 cm **D** 37.1 cm



4 Find the area of this sector. Select **A**, **B**, **C** or **D**.

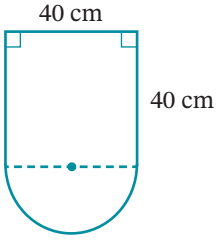
A 19.6 cm^2 **B** 15.3 cm^2

C 8.5 cm^2 **D** 4.4 cm^2

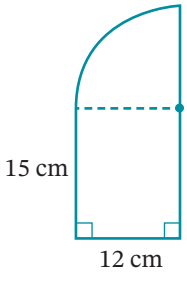


5 Calculate the perimeter of each shape, correct to 2 decimal places.

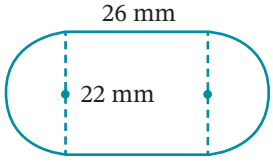
a



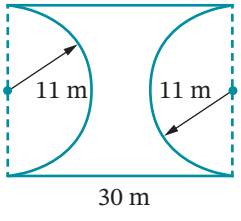
b



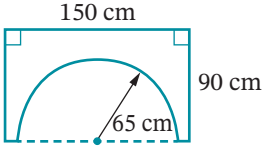
c



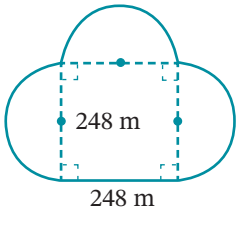
d



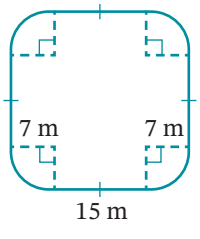
e



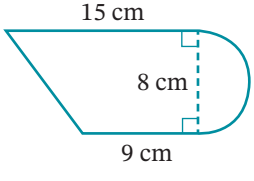
f



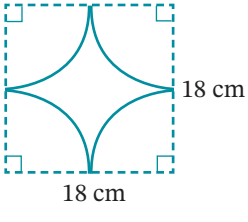
g



h



i



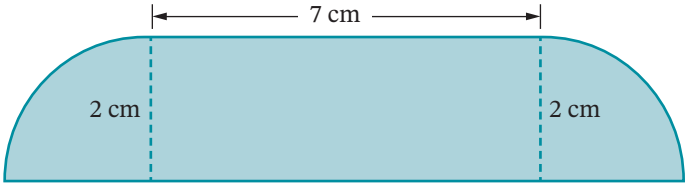
Foundation Standard Complex

9780170479370

EXAMPLE 11

10.06

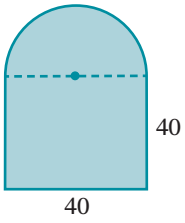
6 Calculate the area of the shape. Select **A**, **B**, **C** or **D**.



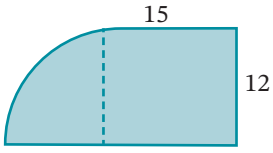
A 26.57 cm^2 **B** 20.28 cm^2 **C** 24.28 cm^2 **D** 28.28 cm^2

7 Calculate the area of each shape, correct to the nearest square metre.
All measurements are in metres.

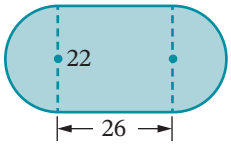
a



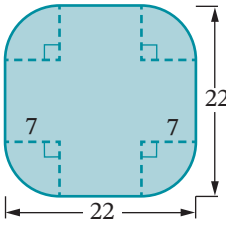
b



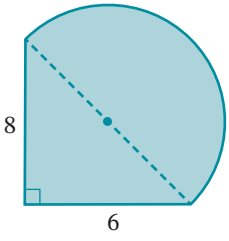
c



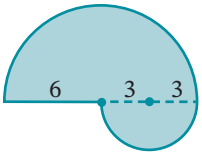
d



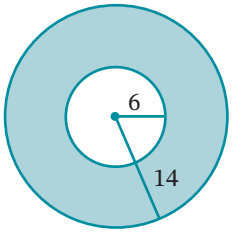
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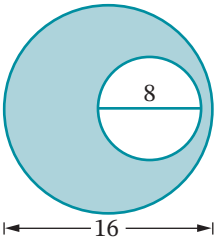
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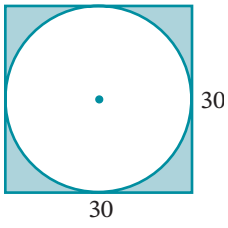
g



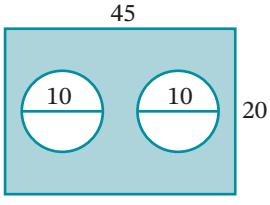
h



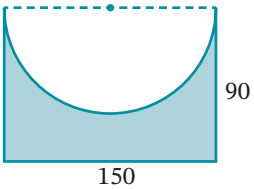
i



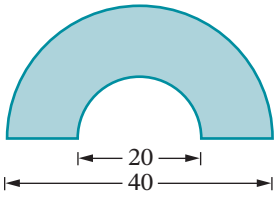
j



k



l



8 A circular playing field has a radius of 80 m. A rectangular cricket pitch measuring 25 m by 2 m is placed in the middle. The field, excluding the pitch, is to be fertilised. PS R

a Calculate to the nearest square metre the area to be fertilised.

b How much will this cost if the fertiliser is $19.95 per 100 square metres? Give your answer to the nearest dollar.

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9 The diameter of the Earth is 12 756 km.

a Find the circumference of the Earth to the nearest kilometre.

b The International Space Station orbits the Earth at an average height of 400 km above the Earth's surface. Find the distance it travels in one orbit of the Earth.



Shutterstock.com/Dima Zel

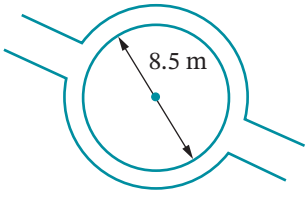
10 A circular plate of diameter 2 m has 250 holes of diameter 10 cm drilled in it. What is the remaining area of the plate? Answer to the nearest 0.1 m^2.

11 A circular pond of diameter 10 m is surrounded by a path one metre wide. PS R

a Calculate the area of the path, correct to 2 decimal places.

b If pavers are $165 per square metre laid, what is the cost of the path?

12 A circular patch of grass has a diameter of 8.5 m. How much further does Arya walk if she walks around the border instead of directly across it? Give your answer correct to the nearest 0.1 m. PS R



13 A new tractor tyre has a diameter of 120 cm, while a worn tyre has a diameter of 115 cm.

a Calculate the difference in circumference between a new and a worn tyre, correct to 3 decimal places.

b Over 1000 revolutions, how much further (to the nearest metre) will a new tyre travel compared to a worn tyre?



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14 A square courtyard measuring 5 m by 5 m has a semi-circular area added to each side. PS R

a Calculate the area of the semi-circular additions, correct to the nearest square metre.

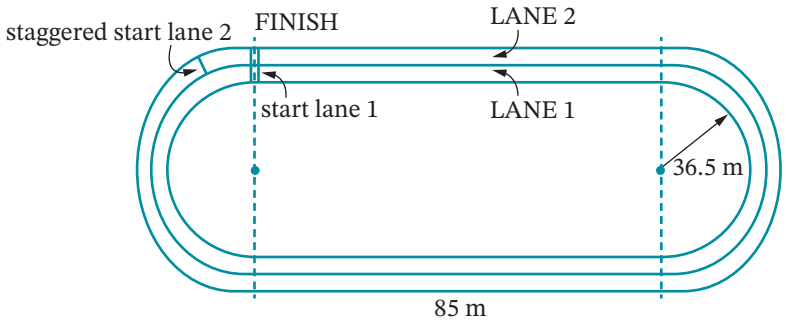
b By what percentage has the area of the courtyard increased?

(This can be calculated as $\frac{\text{increase in area}}{\text{original area}} \times 100\%$)

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15 The measurements of a running track are shown below. Each lane is 1 m wide and the athletes run along the insides of the lanes. PS R



a Izabella runs one lap in lane 1. What distance does she cover, correct to one decimal place?

b Noah runs one lap in lane 2. What distance does he cover, correct to one decimal place?

c By how much should the start line be staggered in lane 2 (from lane 1) so that Izabella and Noah run the same distance in one lap?

16 The wheels of a bicycle have a diameter of 66 cm.

a How far (correct to 2 decimal places) will the bicycle move after one rotation of the wheel?

b How many rotations of the wheel are needed to cover a distance of 5 km?

Did you know?

A piece of pi

$\pi = 3.141\ 592\ 653\ 589\ 793\ 238\ 462\ 643\ 383\ 279\ 502\ 884\ 197\ 169\ 399\ 375\ 105\ 820...$

Because π is an irrational number, its decimal digits run endlessly without repeating. Over history, mathematicians and scientists (and now computer scientists) have tried to calculate more accurate values of π, using more sophisticated formulas and calculation techniques. The ancient Greeks, Romans and Chinese first estimated π as 3. This value is also mentioned in the Bible. Since the first computer, the ENIAC, was invented in 1949, much progress has been made. Supercomputers have been used to calculate more decimal places of π and, in 2022, Emma Haruka Iwao (Japan), working for Google, calculated π to 100 trillion places. In 2015, Suresh Sharma (India) memorised π to 70 030 decimal places. It took him 17 hours 14 minutes to recite it.

Research the modern history of π and find out about recent calculations of its value.

Worksheet
Nets of solids

Investigation

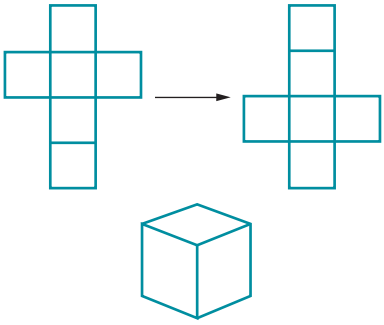
Nets of prisms

You will need: 1 cm grid paper.

If a solid is cut along some of its edges, the faces can all be folded down flat. The flat shape obtained is called the **net** of the solid.

Net of a cube

How many different nets of a cube can you make (or draw)? Do not count reflections or rotations because they are considered to be the same. For example, these are two versions of the same net.



1 Draw the possible nets of a cube on 1 cm grid paper. Cut out the nets and fold them to see if they each form a cube, then paste them into your book.

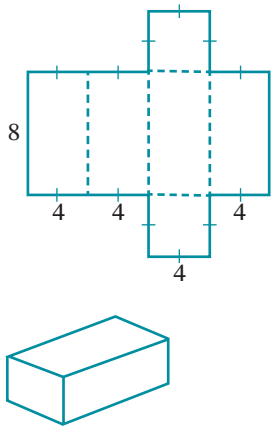
2 If a cube has side length 1 cm, what is the area of one of its faces?

3 The surface area of a cube is the sum of the areas of all of its faces. What is the surface area of a cube with side length 1 cm?

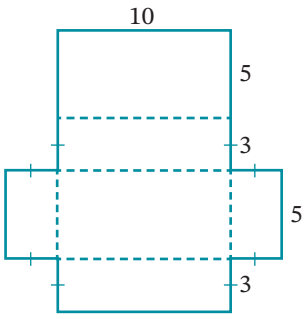
Net of a rectangular prism

1 Draw each of the following nets of rectangular prisms on 1 cm grid paper. Cut them out and fold them to make rectangular prisms, then paste them into your book.

a



b



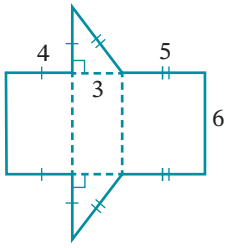
2 Calculate the surface area of each rectangular prism by adding the areas of all of its faces.

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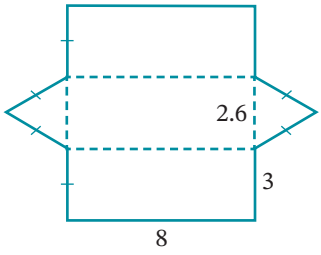
Net of a triangular prism

1 Draw each of the following nets on 1 cm grid paper. Measurements are in centimetres. Cut them out and fold them to make triangular prisms, then paste them into your book.

a



b



2 Calculate the surface area of each triangular prism.

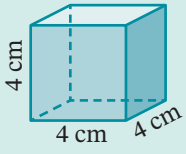
Surface area of a prism 10.07

The **surface area** of a solid is the total area of all the faces of the solid. To calculate the surface area of a solid, find the area of each face and add the areas together.

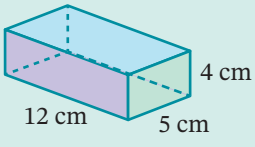
Example 12

Calculate the surface area of each prism.

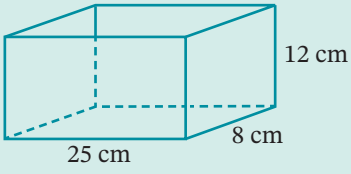
a cube



b rectangular prism



c open rectangular prism



Solution

a A cube has 6 identical faces in the shape of a square.

Surface area $= 6 \times 4^2$ $6 \times$ area of a square

$= 96 \text{ cm}^2$

b A rectangular prism has 6 faces that are not all the same. However, opposite faces, such as the top and bottom, are the same.

Surface area = 2 top and bottom faces + 2 end faces + 2 side faces

$= (2 \times 5 \times 12) + (2 \times 5 \times 4) + (2 \times 12 \times 4)$

$= 256 \text{ cm}^2$

c The open rectangular prism has no top so it has 5 faces that are not all the same.

For the surface area of an open solid, we only count the *external* surfaces, not the internal ones (so that we don't count each surface twice).

Surface area = Bottom face + 2 front and back faces + 2 side faces

$= (25 \times 8) + (2 \times 25 \times 12) + (2 \times 8 \times 12)$

$= 992 \text{ cm}^2$

Worksheets
Surface area 1
Nets of solids

SkillSheet
Solid shapes

Presentation
Surface area

Technology
Surface area calculator

Spreadsheet
Surface area calculator

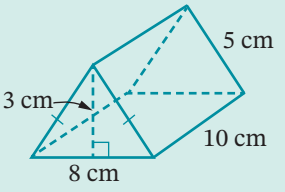
Video
Cubist art

Example 13

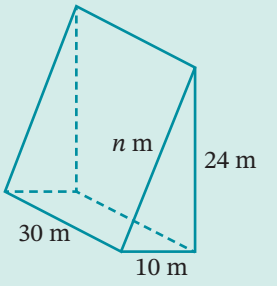
Videos
Surface area of a prism
Surface area of prisms

Calculate the surface area of each triangular prism.

a

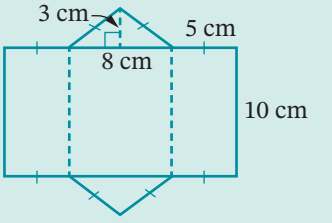


b



Solution

a This triangular prism has 5 faces: 2 identical triangles (front and back), 2 identical rectangles (sides) and another rectangle (bottom). Drawing the net of the prism may help you see this.



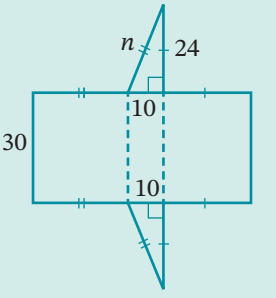
$$\text{Surface area} = (2 \times \frac{1}{2} \times 8 \times 3) + (2 \times 10 \times 5) + (8 \times 10)$$
$$= 204 \text{ cm}^2$$

Front and back faces + side faces + bottom face

b This triangular prism has 5 faces: 2 identical triangles (front and back), 3 different rectangles.

To calculate the area of the left face, we need to know the value of n, which can be found using Pythagoras' theorem.

$$n^2 = 10^2 + 24^2$$
$$= 676$$
$$n = \sqrt{676}$$
$$= 26$$



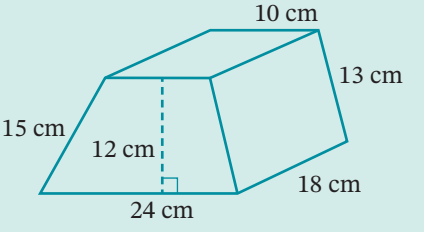
$$\text{Surface area} = (2 \times \frac{1}{2} \times 10 \times 24) + (30 \times 26) + (30 \times 10) + (30 \times 24)$$
$$= 2040 \text{ m}^2$$

Front and back + left + bottom + right

Example 14

Videos
Surface area of a prism

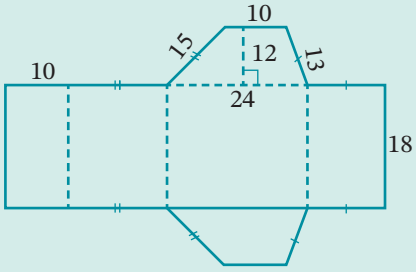
Calculate the surface area of this trapezoidal prism.



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Solution

This trapezoidal prism has 6 faces: 2 identical trapeziums (front and back), 4 different rectangles.



Area of each trapezium $= \frac{1}{2} \times (10 + 24) \times 12$
$= 204 \text{ cm}^2$

Surface area $= (2 \times 204) + (18 \times 10)$
$+ (18 \times 15) + (18 \times 24)$
$+ (18 \times 13)$
$= 1524 \text{ cm}^2$

2 trapeziums + 4 rectangles

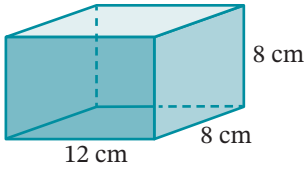
10.07

EXERCISE 10.07 ANSWERS ON P. 589

Surface area of a prism U F PS R

EXAMPLE 12

1 What is the surface area of this rectangular prism? Select the correct answer **A**, **B**, **C** or **D**.



A 320 cm^2 **B** 352 cm^2
C 512 cm^2 **D** 768 cm^2

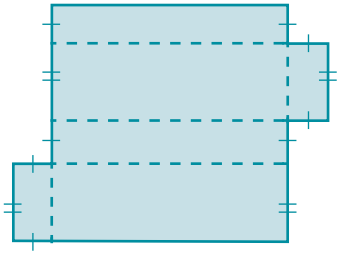
2 Draw a net of each solid.

a cube **b** rectangular prism **c** triangular prism **d** cylinder

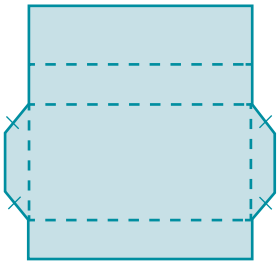
3 Match each net of a solid to the name of the solid.

triangular prism cube rectangular prism
square pyramid cylinder trapezoidal prism

a

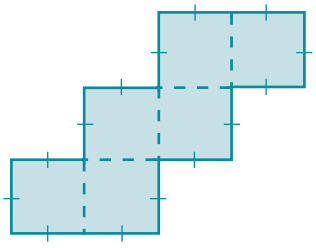


b

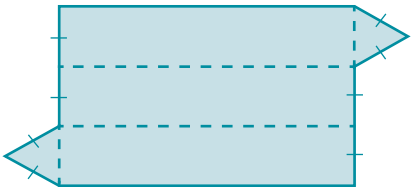


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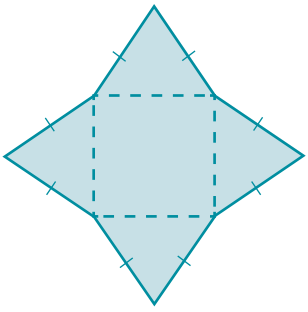
c



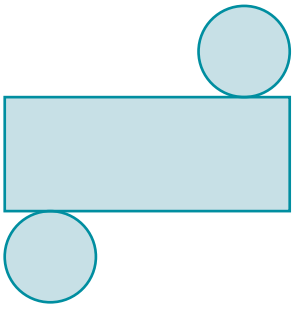
d



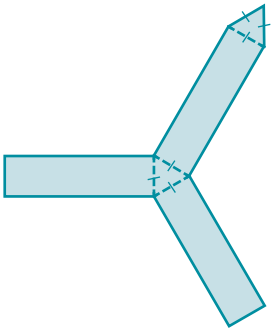
e



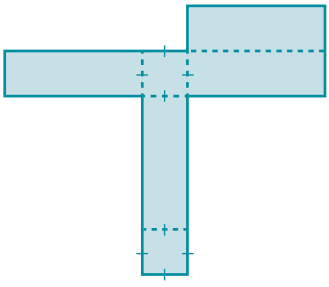
f



g



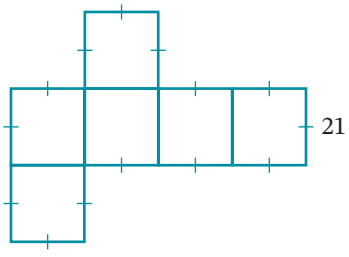
h



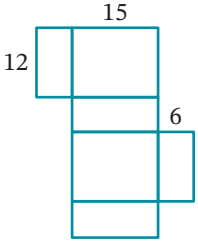
EXAMPLE 13

4 Name the prism that each net represents, then calculate the surface area of the prism. All lengths are in metres.

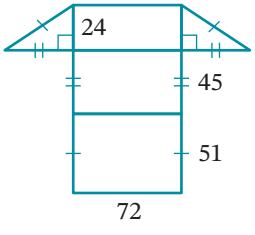
a



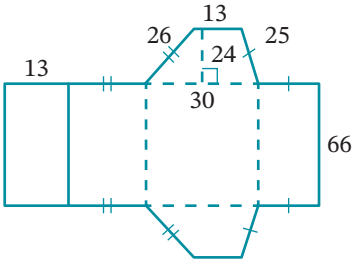
b



c



d

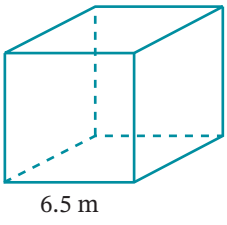


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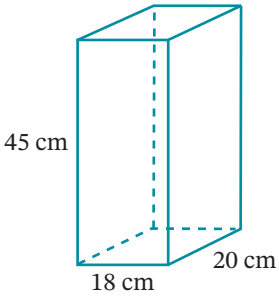
9780170479370

5 Calculate the surface area of each prism. PS R

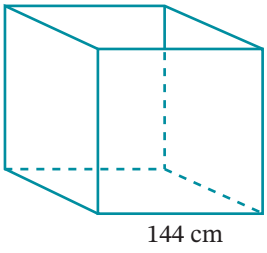
a cube



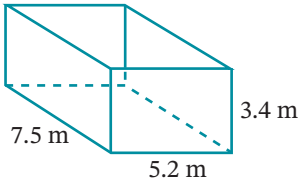
b rectangular prism



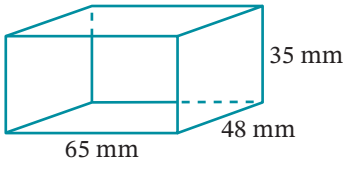
c open cube (no top)



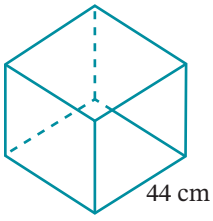
d open rectangular prism



e rectangular prism, open front and back

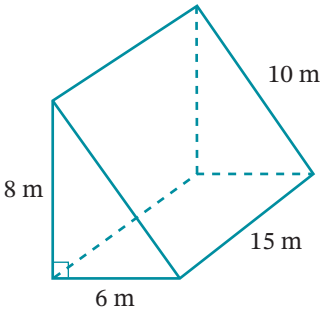


f cube, open one end

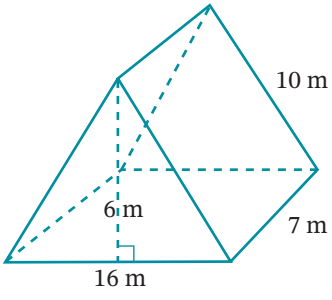


6 Find the surface area of each triangular prism. PS R

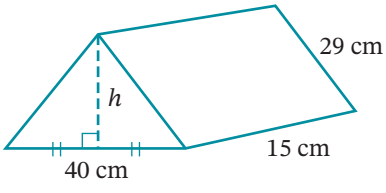
a



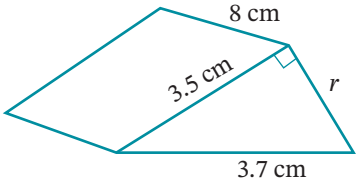
b



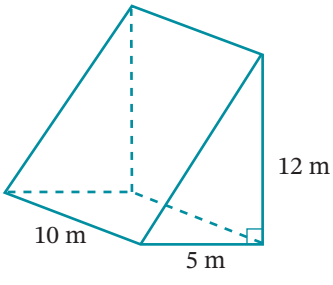
c



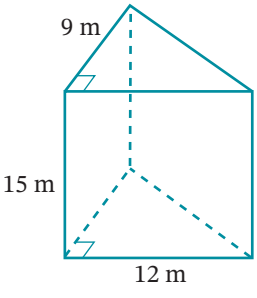
d



e

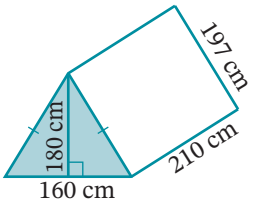


f



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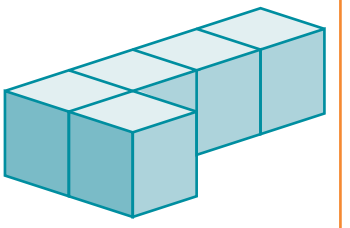
7 Yasmin made this tent. PS R



a Find the surface area of the tent, including the floor.

b Yasmin bought material at $12 per square metre.
How much did it cost her for the material to make the tent?

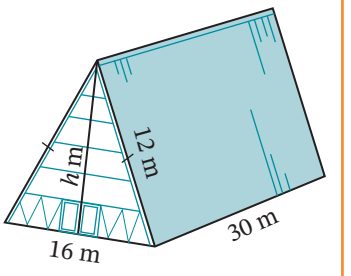
8 5 cubes of side length 1 cm are joined to make this solid.
What is the surface area of the solid? Select **A**, **B**, **C** or **D**.



A 17 cm^2 **B** 18 cm^2

C 5 cm^2 **D** 22 cm^2

9 An art gallery has a width of 16 metres, a length of 30 metres and a slant height of 12 metres, as shown in the diagram. PS R



a Find the perpendicular height, h, of the building, correct to one decimal place.

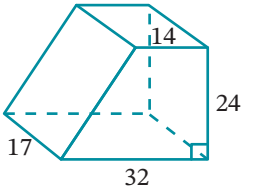
b Find the area of the front wall.

c Find the surface area of the art gallery, without the floor.

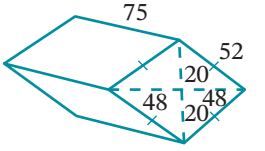
10 Calculate the surface area of each prism. Measurements are in centimetres. PS R

EXAMPLE 14

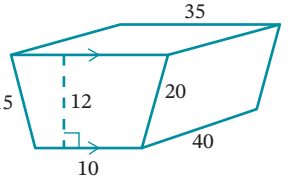
a



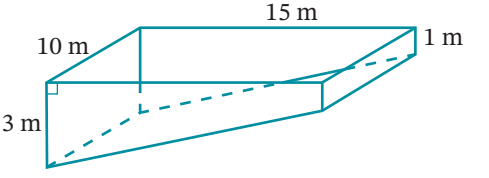
b



c



11 This swimming pool is 15 m long and 10 m wide. The depth of the water ranges from 1 m to 3 m. PS R



Calculate, correct to 2 decimal places:

a the area of the floor of the pool

b the total surface area of the pool.

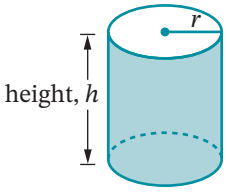
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Investigation

Surface area of a cylinder

Collect 5 different cylindrical cans with paper labels.



1 Copy this table with rows for Cans 1 to 5 and complete it as you perform the following measurements and calculations.

	Circular ends					Curved surface			Surface area
	Height	Diameter	Radius	Area	Circumference	Length	Width	Area	
Can 1									
Can 2									

2 Measure the height and diameter of the first can.

3 Calculate the radius, area and circumference of one circular end, correct to 2 decimal places.

4 Cut the label off the first can and lay it out flat. What shape is this curved surface?

5 Measure the length, width and area of the curved surface (label).

6 Calculate the surface area of the can by adding the areas of the 2 circular ends and the area of the label.

7 Why is the height of each can and the width of its label the same?

8 Why is the circumference of each circular end and the length of the label the same?

9 Repeat Steps **2** to **6** for the other 4 cans.

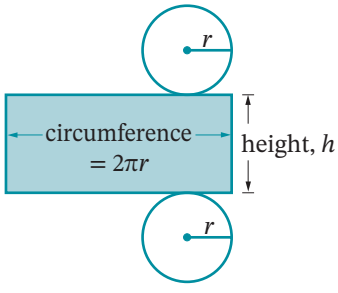
10.08

Surface area of a cylinder

A closed **cylinder** has 3 faces made up of 2 circles (the circular ends) and a rectangle (the curved surface). The length of the rectangle is the circumference of the circular end, while the width of the rectangle is the height of the cylinder.

Surface area of a cylinder = area of 2 circles + area of rectangle

$$SA = 2 \times \pi r^2 + 2\pi r \times h$$
$$= 2\pi r^2 + 2\pi rh$$



Worksheets
Surface area 2
A page of prisms and cylinders
Applications of area 3

Puzzle
Car song

Technology
Surface area calculator

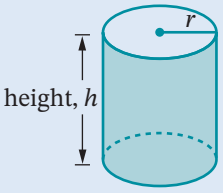
Spreadsheet
Surface area calculator

Surface area of a closed cylinder

$SA = 2\pi r^2 + 2\pi rh$

where r = radius of circular base

h = perpendicular height



The area of the 2 circular ends = $2\pi r^2$ and the area of the curved surface = $2\pi rh$.

Videos
Surface area of a cylinder

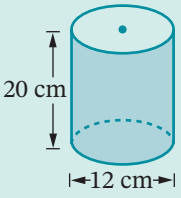
Example 15

Find, correct to one decimal place, the surface area of a cylinder with diameter 12 cm and height 20 cm.

Solution

Radius $= \frac{1}{2} \times 12$ cm $= 6$ cm

$\frac{1}{2}$ of diameter



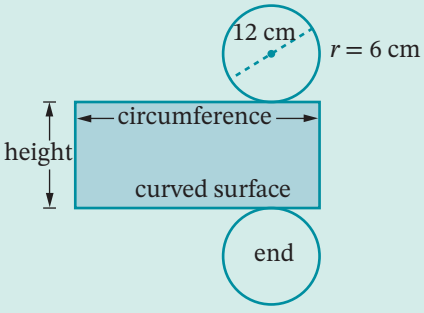
Surface area = area of 2 ends + area of the curved surface

$SA = 2\pi r^2 + 2\pi rh$

$= 2 \times \pi \times 6^2 + 2 \times \pi \times 6 \times 20$

$= 980.1769...$

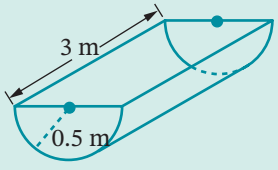
≈ 980.2 cm^2



Example 16

Find, correct to 2 decimal places, the surface area of this open half-cylinder with radius 0.5 m and height 3 m.

Note: For the surface area of an open solid, we only count the external surfaces, not the internal ones (so that we don't count each surface twice).



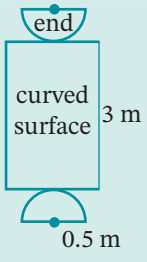
Solution

Surface area = 2 semicircle ends + $\frac{1}{2} \times$ curved surface

$SA = 2 \times (\frac{1}{2} \times \pi \times 0.5^2) + \frac{1}{2} \times (2 \times \pi \times 0.5 \times 3)$

$= 5.49778...$

≈ 5.50 cm^2



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EXERCISE 10.08 ANSWERS ON P. 589

Surface area of a cylinder U F PS R

EXAMPLE 15

1 Calculate, correct to 2 decimal places, the surface area of a cylinder with:

a radius 7 m, height 10 m

b diameter 28 cm, height 15 cm

c diameter 6.2 m, height 7.5 m

d radius 0.8 m, height 2.35 m

10.08

2 Find, correct to one decimal place, the curved surface area of a cylinder with:

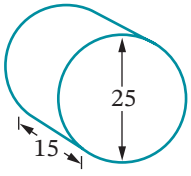
a diameter 9 cm, height 32 cm

b radius 85 mm, height 16 mm

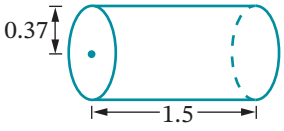
EXAMPLE 16

3 Calculate, correct to one decimal place, the surface area of each solid. All lengths shown are in metres. PS R

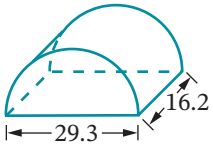
a closed cylinder



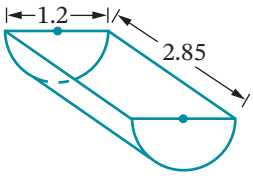
b cylinder with one open end



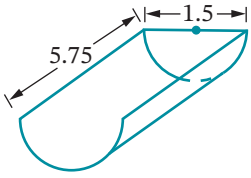
c closed half-cylinder



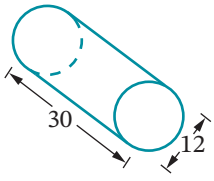
d half-cylinder with open top



e half-cylinder with open top, one end open



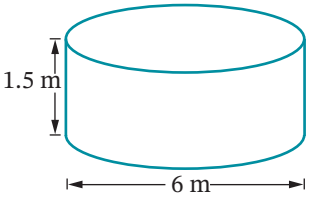
f cylinder open at both ends



4 A swimming pool is in the shape of a cylinder 1.5 m deep and 6 m in diameter. The inside of the pool is to be repainted, including the floor. Find: PS

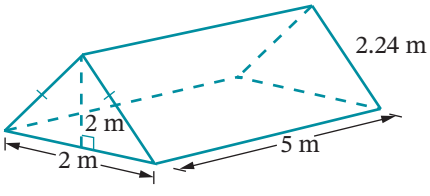
a the area to be repainted, correct to one decimal place

b the number of whole litres of paint needed if coverage is 9 m^2 per litre.

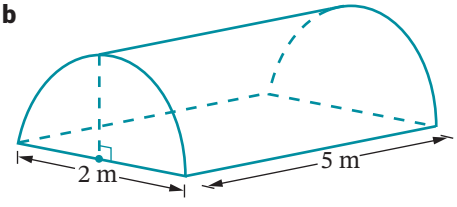


5 Which tent has the greater surface area?

a



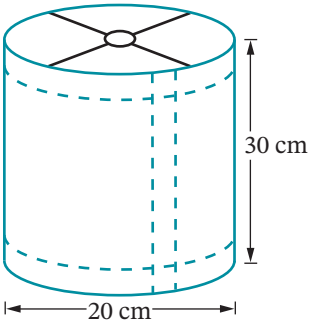
b



(*Note*: the floor is included for both tents)

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6 This lampshade is to be covered. Find the area of material needed, correct to one decimal place, if 10% extra is allowed for seams and overlaps.
Note: The lampshade is a cylinder open at both ends. PS



10.09 Volumes of prisms and cylinders

Worksheets
A page of prisms and cylinders
Volume and capacity

Skillsheet
What is volume?

Puzzle
Formula matching game

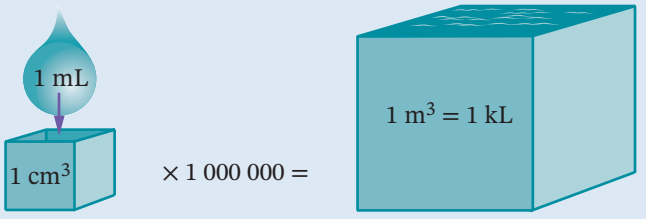
The **volume** of a solid is the amount of space it takes up. Volume is measured in **cubic units**, for example, cubic metres (m^3) or cubic centimetres (cm^3).

The **capacity** of a container is the amount of fluid (liquid or gas) it holds, measured in millilitres (mL), litres (L), kilolitres (kL) and megalitres (ML).

Volume and capacity

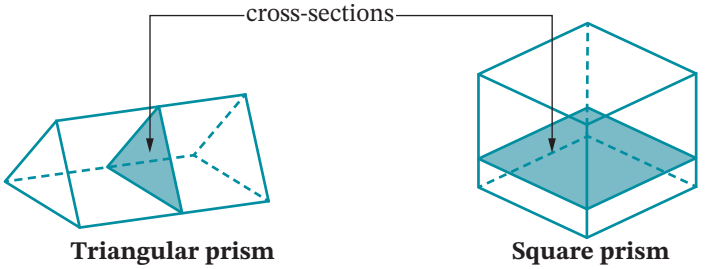
$1\ cm^3$ contains 1 mL

$1\ m^3$ contains 1000 L or 1 kL



Volume of a prism

A **cross-section** of a solid is a 'slice' of the solid cut *across* it, parallel to its end faces, rather than along it. A **prism** has the same (uniform) cross-section along its length, and each cross-section is a **polygon** (with straight sides).

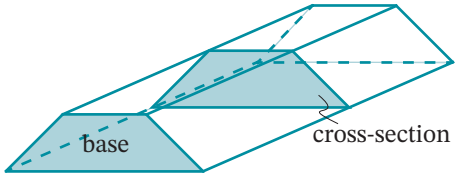


Triangular prism **Square prism**

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An end face of a prism is called its **base**. Prisms take their names from their base and cross-section. For example, this prism is a **trapezoidal prism** because its base and cross-sections are trapeziums.



Trapezoidal prism

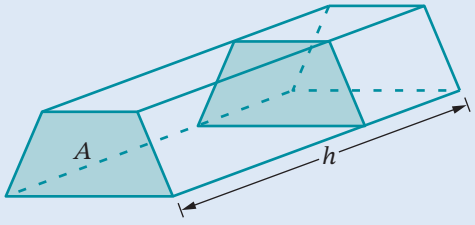
Because a prism is made up of identical cross-sections, its volume can be calculated by multiplying the **area of its base** by its **perpendicular height** (the length or depth of the prism).

Volume of a prism

$V = Ah$

where A = area of base and

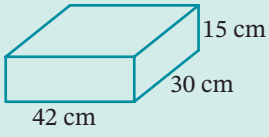
h = perpendicular height



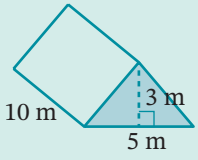
Example 17

Find the volume of each prism.

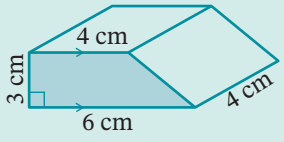
a



b



c



Solution

a $V = 42 \times 30 \times 15$
$= 18\,900 \text{ cm}^3$

For a rectangular prism, volume = length × width × height ($V = lwh$)

b $A = \frac{1}{2} \times 5 \times 3$
$= 7.5$

Area of a triangle.

$V = 7.5 \times 10$
$= 75 \text{ m}^3$

$V = Ah$, where height $h = 10$

c $A = \frac{1}{2} \times (4 + 6) \times 3$
$= 15 \text{ cm}^2$

Area of a trapezium

$V = 15 \times 4$
$= 60 \text{ cm}^3$

$V = Ah$, where height $h = 4$

10.09

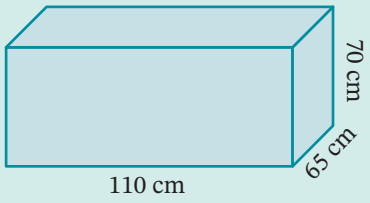
Videos
Volumes of prisms and cylinders

Example 18

A storage box in the shape of a rectangular prism is 110 cm long, 65 cm wide and 70 cm high.

a Find the volume of the box in cm^3.

b What is the volume in m^3?



Solution

a $V = 110 \times 65 \times 70$

$= 500\,500\ cm^3$

b $1\ m = 100\ cm$, so $1\ m^3 = 100^3\ cm^3 = 1\,000\,000\ cm^3$

$\therefore V = 500\,500 \div 1\,000\,000$

$= 0.5005\ m^3$

Alternatively, you could convert 110 cm, 65 cm and 70 cm to metres, then multiply them together for the volume.

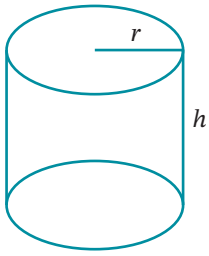
Volume of a cylinder

A cylinder is like a 'circular prism' because its cross-sections are identical circles. Because of this, we can also use $V = Ah$ to find the volume of a cylinder. But for a circle, $A = \pi r^2$, so:

Volume $= Ah$

$V = \pi r^2 \times h$

$= \pi r^2 h$



Volume of a cylinder

$V = \pi r^2 h$

where r = radius of circular base

h = perpendicular height

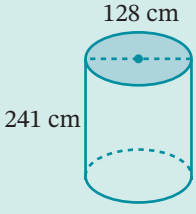
Videos
Volumes of prisms and cylinders

Example 19

For this cylinder, calculate:

a its volume, correct to the nearest cm^3

b its capacity in kL, correct to 1 decimal place.



Solution

a Radius $= \frac{1}{2} \times 128$ cm $= 64$ cm ($\frac{1}{2}$ of diameter)

$V = \pi \times 64^2 \times 241$ ($V = \pi r^2 h$)

$= 3\,101\,179.206...$

$\approx 3\,101\,179\ cm^3$

9780170479370

b Capacity = 3 101 179 mL
= (3 101 179 ÷ 1000 ÷ 1000) kL
= 3.101 179 kL
≈ 3.1 kL

$1\ cm^3 = 1\ mL$



EXERCISE 10.09 ANSWERS ON P. 589

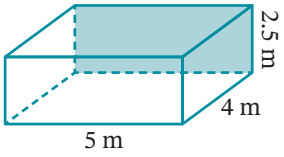
Volumes of prisms and cylinders U F PS R C

10.09

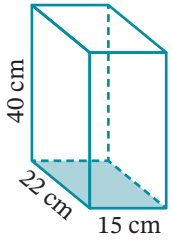
EXAMPLE 17

1 Find the volume of each rectangular prism using the formula $V = lwh$.

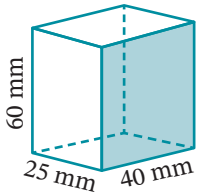
a



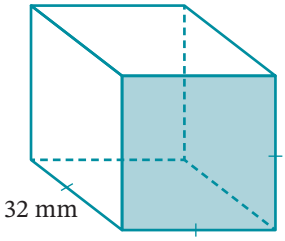
b



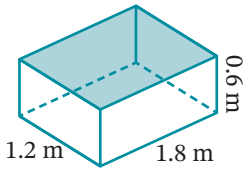
c



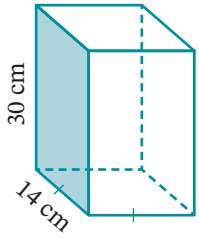
d



e



f



EXAMPLE 18

2 The volume of a cube is 125 000 cm^3. Convert this to m^3. Select the correct answer **A**, **B**, **C** or **D**.

A 1250 m^3 **B** 12.5 m^3 **C** 1.25 m^3 **D** 0.125 m^3

3 The volume of a small jewellery box is 160 000 mm^3. Which expression gives its volume in cm^3? Select **A**, **B**, **C** or **D**.

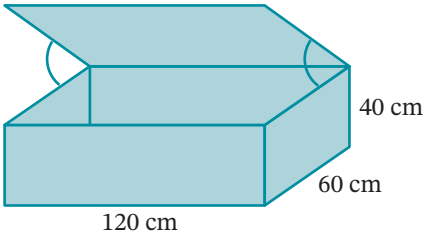
A $160\,000 \div 10^3$ **B** $160\,000 \div 100$
C $160\,000 \div 100^2$ **D** $160\,000 \div 10^2$

4 Find the volume of this toolbox in cubic metres by: PS R C

a calculating the volume in cm^3, then converting to m^3

b converting each length to metres first, then calculating the volume.

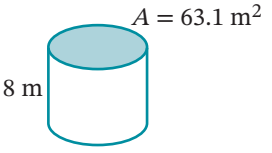
Which method is easier?



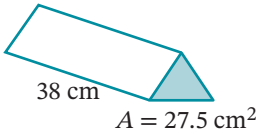
Foundation Standard Complex

5 Find the volume of each solid.

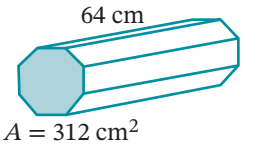
a



b



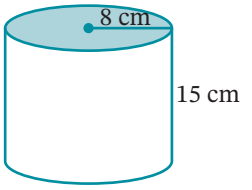
c



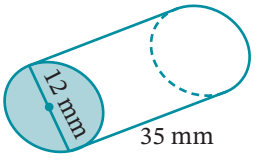
EXAMPLE 19

6 Find, correct to nearest whole unit, the volume of each cylinder.

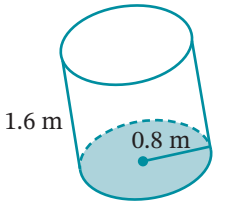
a



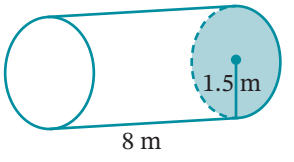
b



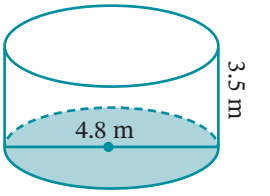
c



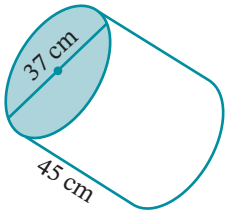
d



e

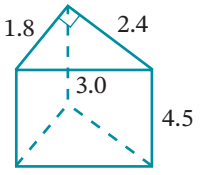


f

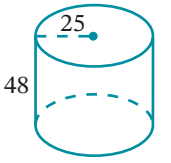


7 Calculate, correct to one decimal place, the volume of each solid. All lengths are in metres.

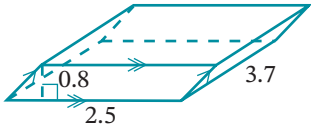
a



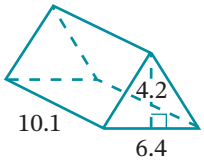
b



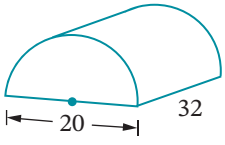
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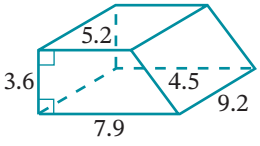
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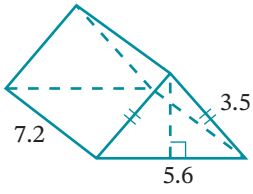
e



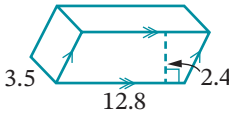
f



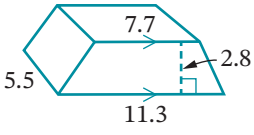
g



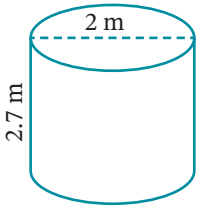
h



i



8 What is the capacity of this cylinder in litres? Select the closest answer **A**, **B**, **C** or **D**.



A 33 900 L **B** 8482 L

C 6280 L **D** 16 964 L

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9 A triangular prism has base length 15 cm, height 8 cm and volume 630 cm^3. What is its length? R

10 A fish tank that is 55 cm long, 24 cm wide and 22 cm high is filled to 4 cm below the top. Calculate the amount of water in the tank in litres.

11 A wedding cake with 3 tiers rests on a table. Each tier is 6 cm high. The layers have radii of 20 cm, 15 cm and 10 cm respectively. Find the total volume of the cake, correct to the nearest cm^3.



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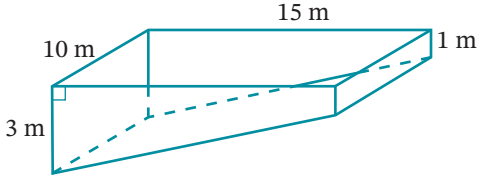
12 Calculate the volume of the skip bin, correct to the nearest m^3.



Shutterstock.com/Stephen Rees

13 This swimming pool is 15 m long and 10 m wide. The depth of the water ranges from 1 m to 3 m.

Calculate the capacity of this pool in kilolitres.



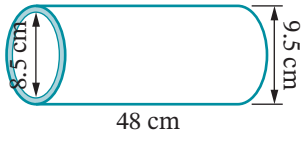
14 Robyn and Anthony are planning to make a raised vegetable garden. If the dimensions of the garden bed are 7.5 m × 9 m × 60 cm, calculate how much soil they will need to buy. Select **A**, **B**, **C** or **D**.

A 405 m^3 **B** 40.5 m^3

C 4050 cm^3 **D** 4050 m^3


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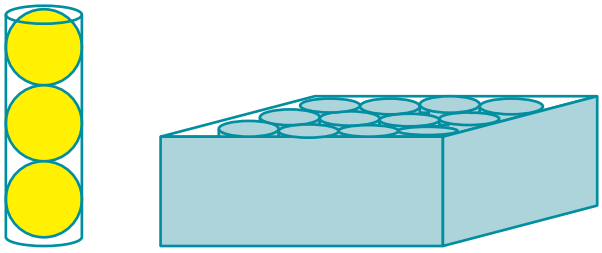
15 This metal pipe has an inner diameter of 8.5 cm and an outer diameter of 9.5 cm. Calculate, correct to 2 decimal places, the volume of metal needed to make the pipe. PS R



16 A cube has a volume of 2150 mm^3. What is its side length, correct to the nearest millimetre? R

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17 The official diameter of a tennis ball is defined as 6.54 cm to 6.86 cm. PS R C



a Find the average diameter of a tennis ball.

b The tennis balls are packed in cylinders as shown. Using the larger diameter, calculate the radius and height of the cylinders.

c **i** To find the volume of the cylinder, its radius and height are rounded *up* to one decimal place. Explain why, and write the radius and height rounded up.

ii Hence, calculate the volume of a cylinder, correct to one decimal place.

d For packaging, 12 cylinders are put in a cardboard box, as shown above.

i What are the dimensions of the box?

ii Calculate the volume of the box.

e Calculate the volume of the box not taken up by the cylinders.

f What percentage (correct to one decimal place) of the box is space not taken up by the cylinders?

Investigation

Volume vs surface area

You will need: At least 20 centicubes

1 Copy the following table.

Length	Width	Height	Volume	Surface area
			8	
			8	

2 Build as many rectangular prisms as you can with a volume of 8 cubes, but with different dimensions. *Note*: $1 \times 1 \times 8$ is the same as $1 \times 8 \times 1$ and $8 \times 1 \times 1$.

3 Record your dimensions in the table and calculate the surface area of each prism.

4 What are the dimensions of the solid that has the smallest surface area?

5 What name do we give to this solid?

6 Repeat the experiment for a volume of:

a 12 cubes **b** 20 cubes.

Mental skills 10B: Maths without calculators ANSWERS ON P. 590

Quiz
Mental skills 10B

Finding 15%, $2\frac{1}{2}$%, 25% and $12\frac{1}{2}$%

- To find 10% or $\frac{1}{10}$ of a number, divide by 10.
- To find 5% of a number, find 10% first, then halve it (since 5% is half of 10%).
- So to find 15% of a number, find 10% and 5% of the number separately, then add the answers together.

1 Study each example.

a $15\% \times 80 = (10\% \times 80) + (5\% \times 80)$
$= 8 + 4$
$= 12$

b $15\% \times \$170 = (10\% \times \$170) + (5\% \times \$170)$
$= \$17 + \$8.50$
$= \$25.50$

c $15\% \times 3600 = (10\% \times 3600) + (5\% \times 3600)$
$= 360 + 180$
$= 540$

d $15\% \times \$28 = (10\% \times \$28) + (5\% \times \$28)$
$= \$2.80 + \$1.40$
$= \$4.20$

2 Now find 15% of each amount.

a	120	**b**	\$840	**c**	260	**d**	\$202
e	\$50	**f**	72	**g**	\$180	**h**	400
i	\$1600	**j**	\$22	**k**	6000	**l**	\$350

To find $2\frac{1}{2}$% of a number, first find 5%, then halve it.

3 Study each example.

a $2\frac{1}{2}\% \times 600$
$10\% \times 600 = 60$
$5\% \times 600 = \frac{1}{2} \times 60 = 30$
$2\frac{1}{2}\% \times 600 = \frac{1}{2} \times 30 = 15$

b $2\frac{1}{2}\% \times \$820$
$10\% \times \$820 = \$82$
$5\% \times \$820 = \frac{1}{2} \times 82 = \$41$
$2\frac{1}{2}\% \times \$820 = \frac{1}{2} \times \$41 = \$20.50$

4 Now find $2\frac{1}{2}$% of each amount.

a	400	**b**	6640	**c**	\$2000	**d**	\$880
e	1500	**f**	\$232	**g**	5400	**h**	\$904

To find 25% of a number, halve the number twice as $25\% = \frac{1}{4}$.

5 Study each example.

a $25\% \times 700$
$50\% \times 700 = \frac{1}{2} \times 700 = 350$
$25\% \times 700 = \frac{1}{2} \times 350 = 175$

b $25\% \times \$86$
$50\% \times \$86 = \frac{1}{2} \times \$86 = \$43$
$\therefore 25\% \times \$86 = \frac{1}{2} \times \$43 = \$21.50$

6 Now find 25% of each of each amount.

a	2000	**b**	\$80	**c**	18	**d**	\$25
e	\$324	**f**	\$140	**g**	66	**h**	298
i	\$780	**j**	\$1700	**k**	\$126	**l**	1160

To find $12\frac{1}{2}\%$ of a number, find 25% first, then halve it. In other words, halve three times because $12\frac{1}{2}\% = \frac{1}{8}$.

7 Study each example.

a $12\frac{1}{2}\% \times 400$

$50\% \times 400 = \frac{1}{2} \times 400 = 200$

$25\% \times 400 = \frac{1}{2} \times 200 = 100$

$12\frac{1}{2}\% \times 400 = \frac{1}{2} \times 100 = 50$

b $12\frac{1}{2}\% \times \$144$

$50\% \times \$144 = \frac{1}{2} \times \$144 = \$72$

$25\% \times \$144 = \frac{1}{2} \times \$72 = \$36$

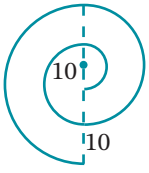
$12\frac{1}{2}\% \times \$144 = \frac{1}{2} \times \$36 = \$18$

8 Now find $12\frac{1}{2}\%$ of each amount.

a	1280	**b**	\$12	**c**	60	**d**	\$260
e	\$540	**f**	\$250	**g**	304	**h**	1360

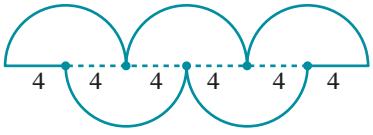
Power plus ANSWERS ON P. 590

1 A spiral is formed from 4 semicircles as shown. The diameter of the smallest semicircle is 10 cm, and the semicircles are 10 cm apart. Find the total length of the spiral:



a correct to 2 decimal places **b** in terms of π.

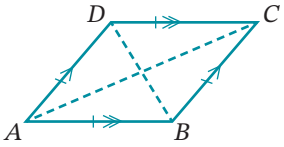
2 In this pattern, all lengths are in metres. Calculate, correct to 2 decimal places:



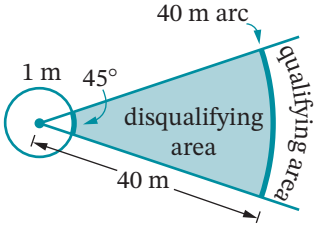
a its perimeter

b its area

3 A rhombus has diagonals of length 24 cm and 18 cm. Find its perimeter.



4 To qualify for the next round of the discus trials, Bronte must throw the discus beyond the 40 m arc. Find, correct to the nearest square metre, the shaded disqualifying area of the sector, given that the small circle has a radius of 1 m.



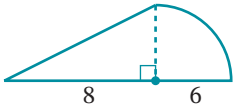
 9780170479370

5 For each composite shape calculate, correct to one decimal place:

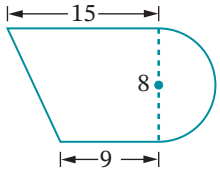
i its area **ii** its perimeter.

All measurements are in centimetres.

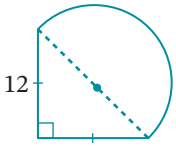
a



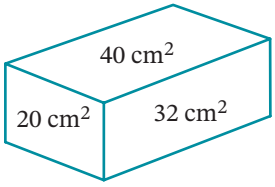
b



c

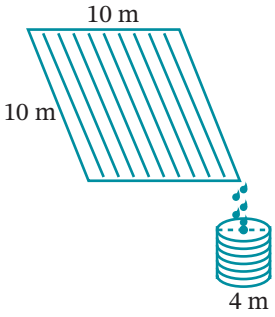


6 The 3 faces of a rectangular prism have areas as shown. Calculate the volume of the prism.



7 A cube opened at one end has an external surface area of 1125 cm^2. Find its volume.

8 A 10 m flat square roof drains into a cylindrical rainwater tank with a diameter of 4 m. If 5 mm of rain falls on the roof, by how much (to the nearest mm) does the level of water in the tank rise?



10.09

CHAPTER 10 REVIEW

Language of maths

Quiz
Language of maths 10

Puzzle
Surface area and volume crossword

absolute error	arc length	base	capacity
circumference	cross-section	curved surface	diagonal
diameter	giga-	micro- (μ)	net
percentage error	perimeter	perpendicular height	pi (π)
precision	prefix	radius/radii	rectangular prism
sector	surface area	triangular prism	

1 Draw a circle and label on it:

a a sector **b** a diameter **c** an arc length

2 In the formula $V = Ah$, explain what V, A and h stand for.

3 Which metric prefix means 'one-millionth'?

4 Write the definition of a **prism** using the word **cross-section**.

5 How do you find the **surface area** of a solid?

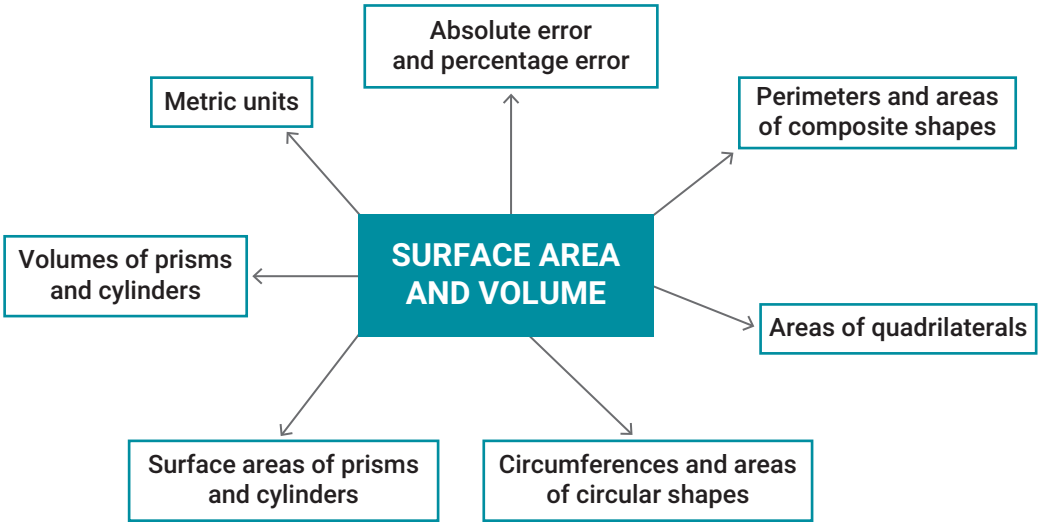
6 What is the **precision** of your ruler?

Topic summary

Worksheet
Mind map: Surface area and volume

- Write 10 questions (with solutions) that could be used in a test for this chapter. Include some questions that you have found difficult to answer.
- Swap your questions with another student and check their solutions against yours.
- List the sections of work in this chapter that you did not understand. Follow up this work with your friend or teacher.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



TEST YOURSELF 10

ANSWERS ON P. 590

1 Convert: 10.01

a 8 km to m **b** 15 min to s **c** 0.7 m to mm
d 250 000 g to kg **e** 8.4 GB to MB **f** 9.5 t to kg
g 300 min to h **h** 34 000 B to kB **i** 125 kL to L
j 2500 cm to km **k** 17 000 000 kg to t **l** 8900 GB to TB
m 10 500 ms to s **n** 9.1 millennia to years **o** 2.6 Mm to m
p 3 ly to km **q** 0.000 000 4 s to μs **r** 75 000 000 μm to m

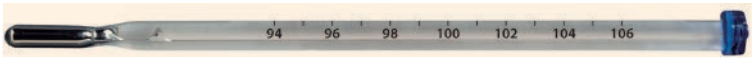
2 For each measuring instrument, state: 10.02

i its precision **ii** its absolute error.

a



b

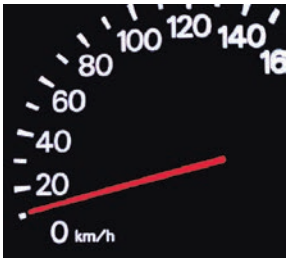


c



Shutterstock.com/Sergey Melnikov

d

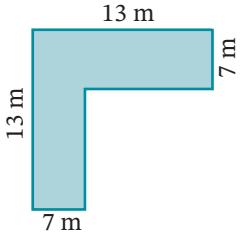


3 A super yacht is 24 m long. For this measurement, calculate: 10.03

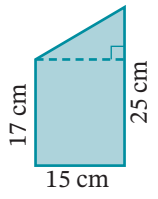
a its absolute error **b** its percentage error correct to 2 decimal places

4 Calculate the perimeter of each shape. 10.04

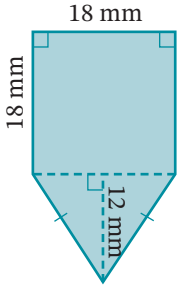
a



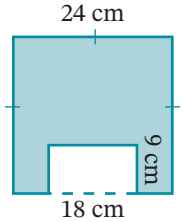
b



c



d

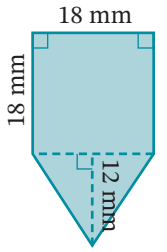


Foundation | Standard | Complex

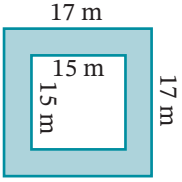
10.04

5 Calculate the area of each shape.

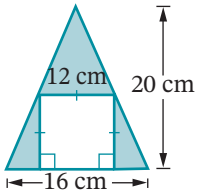
a



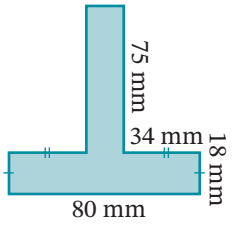
b



c



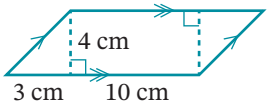
d



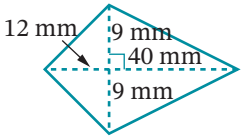
10.05

6 Calculate the area of each quadrilateral.

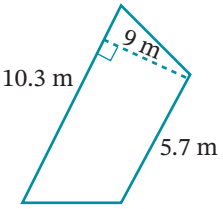
a



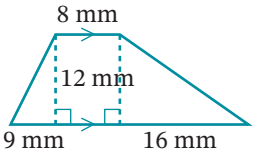
b



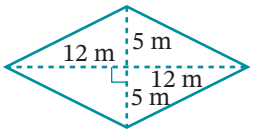
c



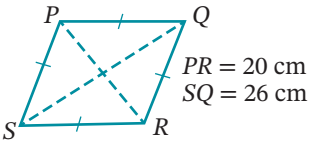
d



e



f

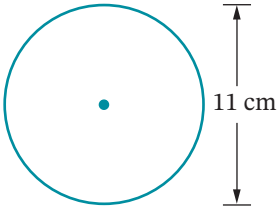


10.06

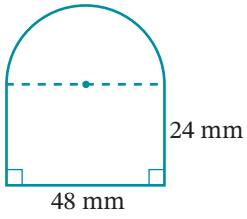
7 For each shape calculate, correct to one decimal place:

i its perimeter **ii** its area.

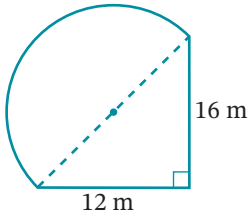
a



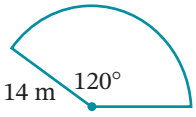
b



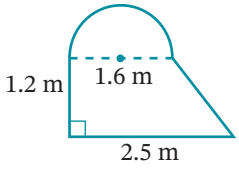
c



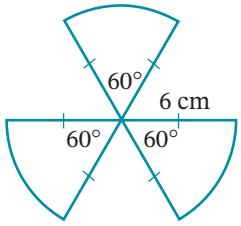
d



e



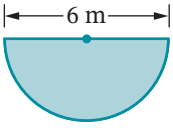
f



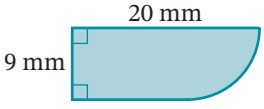
10.06

8 Calculate, correct to 2 decimal places, the area of each shape.

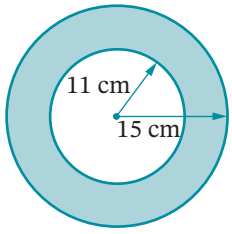
a



b



c

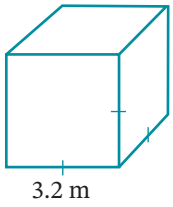


Foundation Standard Complex

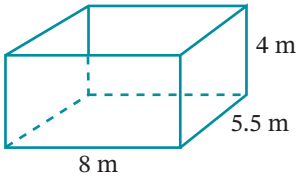
9780170479370

9 Calculate the surface area of each prism. 10.07

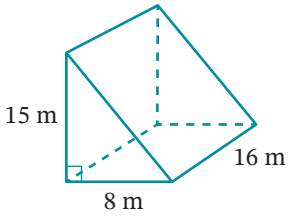
a closed cube



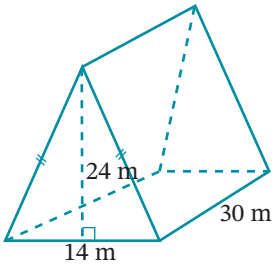
b open rectangular prism



c



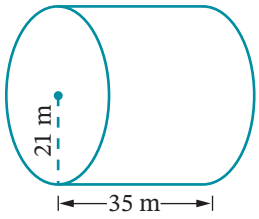
d



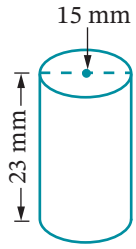
10 Calculate the volume of each prism in question **8**. 10.08

11 Calculate, correct to 2 decimal places, the surface area of each cylinder. 10.08

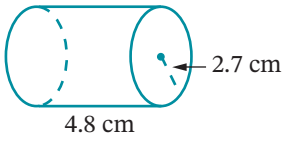
a



b



c



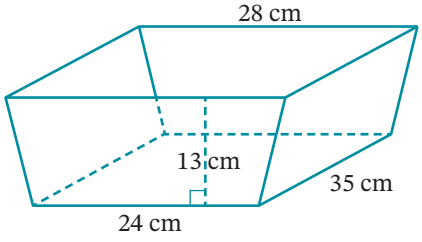
Cylinder, open at one end

12 Calculate, correct to 2 decimal places, the volume of each cylinder in question **10**. 10.09

13 A cylindrical pipe (open at both ends) is 50 cm long and has a diameter of 15 cm. Calculate its external surface area correct to the nearest square centimetre. 10.08

14 A rectangular fish tank that is 75 cm long by 55 cm wide by 35 cm deep is filled to 5 cm from the top. How many litres of water is in the tank? 10.09

15 Calculate the volume of this lunchbox. 10.09



Foundation | Standard | Complex

PRACTICE SET 3

ANSWERS ON P. 590

10.01

1 Copy and complete:

a 750 min = _____ h

b 1350 mL = _____ L

c 512 MB = _____ GB

d 3.005 kg = _____ g

e 2.7 km = _____ m

f 0.25 MB = _____ kB

8.03

2 Mason is paid $3.94 for each toy he assembles.

a How much does he earn for assembling 140 toys?

b How many toys did Mason assemble if he earned $1260.80?

8.04

3 Katrina's annual salary is $85 000 and she earns another $4800 from investments. Katrina's work-related tax deductions amount to $2380. Calculate Katrina's:

a taxable income

b income tax payable using the table below.

Taxable income	Tax on this income
0–$18 200	Nil
$18 201–$45 000	19% for each $1 over $18 200
$45 001–$120 000	$5092 plus 32.5c for each $1 over $45 000
$120 001–$180 000	$24 467 plus 37c for each $1 over $120 000
$180 001 and over	$51 667 plus 45c for each $1 over $180 000

Source: ATO

9.02

4 The shoe sizes of 30 Year 9 students are:

6	10	7	6	8	6	7	7	6	5
5	6	5	8	10	11	9	8	9	5
6	7	9	8	7	6	5	6	7	8

a Arrange the data in a frequency table.

b Draw a frequency histogram and polygon of this data.

c What percentage (correct to one decimal place) of students had shoe sizes less than 8?

d Find the mode and the range.

e Find the median.

f Calculate the mean, correct to one decimal place.

8.01

5 For a salary of $89 760 p.a., calculate, correct to the nearest cent, the:

a weekly income **b** fortnightly income **c** monthly income.

8.03

6 Bhavin works 38 hours a week and is paid $43.50 per hour. Calculate:

a Bhavin's weekly income

b the holiday loading of $17\frac{1}{2}$% on 4 weeks' pay

c Bhavin's fortnightly pay when he receives his holiday loading.

Foundation Standard Complex

9780170479370

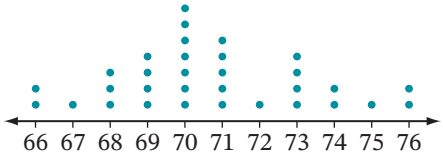
7 For each set of data, find: 9.01

i the mode **ii** the mean (correct to one decimal place)

iii the median **iv** the range

a 47 46 50 51 48 48 52 46 49

b



8 Classify whether each type of data is categorical or numerical. If categorical, then classify whether it is nominal or ordinal. If numerical, then classify whether it is discrete or continuous. 9.05

a the age Year 9 students turn this year **b** colour of cars in a car park

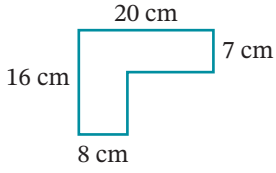
c travelling time between home and work **d** amount of rainfall

e shoe size **f** film classification (e.g. PG, M)

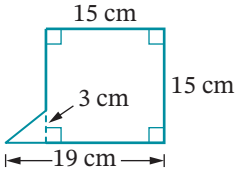
9 For each composite shape, find: 10.03

i its perimeter **ii** its area

a



b



10 Nadia's gross pay for a fortnight is $3038. She pays $724.50 in tax and her other deductions amount to $285. 8.05

a Calculate Nadia's net pay.

b Express her net pay as a percentage (correct to one decimal place) of her gross pay.

11 For the data in this stem-and-leaf plot, which statement is true? Select the correct answer **A**, **B**, **C** or **D**. 9.02

Stem	Leaf
6	0 1 5
7	3 4 4 8 9
8	0 1 2 2 5 6 6 6
9	3 4 7
10	6

A The mean is 82.

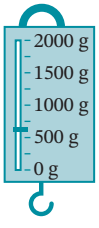
B The mode is 6.

C The range is 46.

D The median is 81.

12 Sefa is paid $27.40 per hour. Calculate his earnings for working 38 hours at normal time, 8 hours at time-and-a-half and 4 hours at double time. 8.02

13 What is the absolute error for these measuring scales? 10.02



14 The diameter of a ball is 8 cm, measured to the nearest centimetre. 10.03

a What is the absolute error of this measurement?

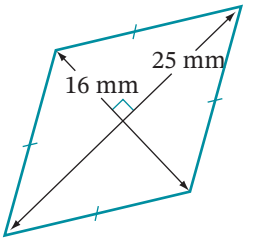
b Calculate its percentage error.

Foundation Standard Complex

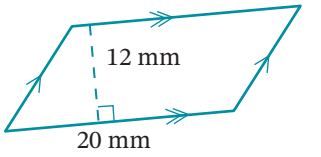
10.04

15 Find the area of each quadrilateral.

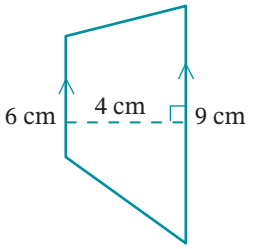
a



b



c



8.03

16 Thiago earns a commission of 2.75% on every property he sells. Calculate his commission for selling an apartment for $635 200.

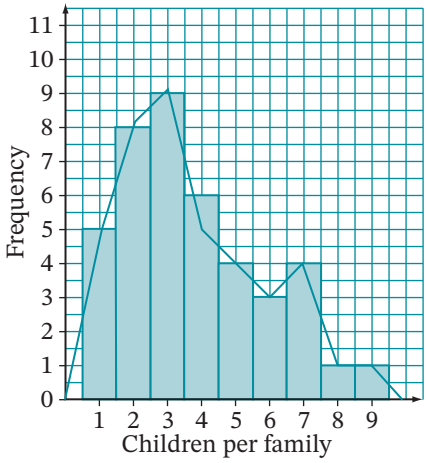
9.03

17 A sample of students was surveyed on the number of children in their families, and the results were graphed on a frequency histogram and polygon.

a Describe the shape of the distribution.

b Where does clustering occur?

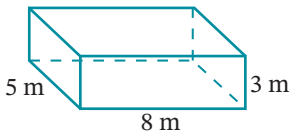
c Find the mode and range.



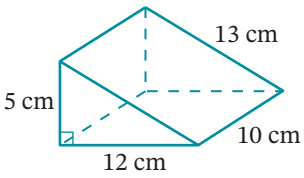
10.07

18 Find the surface area of each prism.

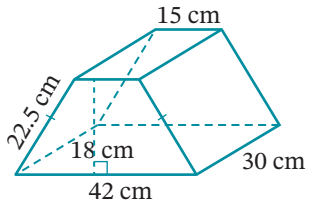
a



b



c



10.09

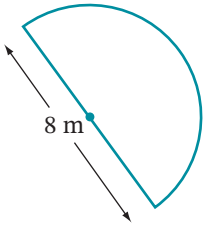
19 Find the volume of each prism shown in question **17**.

10.06

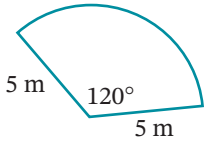
20 For each circular shape, calculate, correct to one decimal place:

i its perimeter **ii** its area.

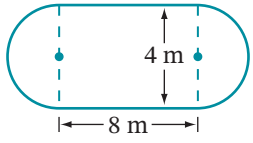
a



b



c



Foundation Standard Complex

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PATH 9.06

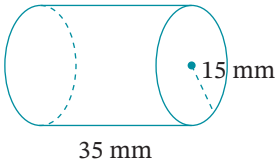
21 In a survey, a sample of people at a park were asked 'Are you employed?'

a Why might this sample of people be biased?

b What is wrong with the survey question?

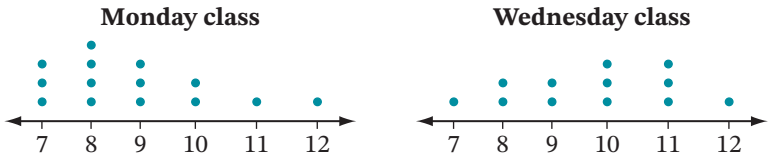
c Suggest a better-worded question.

22 Find, correct to 3 significant figures, the volume of this cylinder. 10.09



23 Find, correct to 3 significant figures, the surface area of the cylinder in question **21** if it is open at one end. 10.08

24 These dot plots show the ages of students in 2 Taekwondo classes. 9.04



a Describe the shape of the age distribution for each class.

b Find the mean for each class.

c Find the median for each class.

d Which class had more consistent ages?

e Which class had more older students?

25 A cube has a surface area of 614 cm^2. What is the length of one side of the cube, correct to one decimal place? 10.07

26 Zahara works a 38-hour week and her hourly pay rate is $28.70. Calculate Zahara's fortnightly pay. 8.01

27 The hourly pay rate for people working at a catering and party hire company are: 9.01

$25.80	$34.00	$27.60	$27.60	$32.50	$48.90
$32.50	$34.70	$25.80	$35.40	$38.10	$32.50

a Find the mean and median pay rate.

b Which pay rate is the outlier?

c Find the mean and median without the outlier.

d How does the outlier affect the mean and median? Explain.

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